

GUJRAAT WATER SUPPLY & SEWERAGE BOARD

(GWSSB - A Government of Gujraat Undertaking)

TENDER DOCUMENT Design, Build and Operate (DBO) of a 100 MLD Conventional Water Treatment Plant with Raw Water Pumping, Clear Water Pumping, Rising Main, and 10-Year Operation & Maintenance, at Surat

Field	Details
Tender Reference No.	GWSSB/WTP-SURAT/2026/DBO-009
Issuing Authority	Gujraat Water Supply & Sewerage Board (GWSSB)
Funding	State Plan Funds + AMRUT 2.0
Mode of Tendering	Two-Envelope System (Technical Bid + Price Bid), e-Procurement Portal
Tender Fee	₹25,000 (Non-refundable)
Earnest Money Deposit (EMD)	₹2,00,00,000 (Refundable, or 2% of quoted D&B Bid Value, whichever is higher)
Bid Validity	180 days from the date of Price Bid opening
Date of Issue	10-Jan-2026
Pre-Bid Meeting	22-Jan-2026, 11:00 AM, GWSSB Circle Office, Surat
Last Date for Submission	05-Mar-2026, 5:00 PM (IST)
Technical Bid Opening	06-Mar-2026, 11:00 AM (IST)
Price Bid Opening	Intimated to technically qualified bidders only
Contract Type	Design-Build (D&B) Lump Sum + 10-Year O&M (Fixed + Variable Fee)

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PART A - Instructions to Bidders (ITB)

This Part sets out the procedures to be followed by prospective bidders in the preparation and submission of their bids, and by the Employer in the receipt, opening, evaluation of bids and award of Contract.

ITB-1: Scope of Bid

The Gujraat Water Supply & Sewerage Board (GWSSB), hereinafter referred to as "the Employer", invites bids from eligible, experienced and financially sound bidders for the Design, Build and Operate (DBO) of a 100 MLD Conventional Water Treatment Plant at Surat, together with associated raw water pumping, clear water pumping, rising main, and a 10-year comprehensive Operation & Maintenance (O&M) contract, as more particularly described in Part D (Scope of Work) and Part E (Technical Specifications) of this Tender Document.

Bidders should note that non-compliance with this clause may render the bid liable for rejection at the sole discretion of the Employer, and bidders are advised to seek clarification well in advance of the bid due date if any ambiguity is perceived in the requirements of this clause, through the process described under ITB-7 (Clarification of Bidding Documents).

ITB-2: Source of Funds

The Project is being funded through a combination of State Plan Funds and financial assistance under the Atal Mission for Rejuvenation and Urban Transformation (AMRUT 2.0), and bidders shall note that all payments under the resulting Contract are subject to the timely release of funds under this financing arrangement, though this shall not relieve the Employer of its payment obligations under the Contract.

Bidders should note that non-compliance with this clause may render the bid liable for rejection at the sole discretion of the Employer, and bidders are advised to seek clarification well in advance of the bid due date if any ambiguity is perceived in the requirements of this clause, through the process described under ITB-7 (Clarification of Bidding Documents).

ITB-3: Eligible Bidders

This Tender is open to single entities and Joint Ventures (JV) of up to two members meeting the eligibility criteria set out in Part F. In case of a JV, the Lead Member shall hold a minimum of 51% equity stake and shall independently meet at least 60% of the minimum eligibility criteria for average annual turnover and relevant experience.

Bidders should note that non-compliance with this clause may render the bid liable for rejection at the sole discretion of the Employer, and bidders are advised to seek clarification well in advance of the bid due date if any ambiguity is perceived in the requirements of this clause, through the process described under ITB-7 (Clarification of Bidding Documents).

ITB-4: Cost of Bidding

The bidder shall bear all costs associated with the preparation and submission of its bid, and the Employer shall in no case be responsible or liable for those costs, regardless of the outcome of the bidding process.

Bidders should note that non-compliance with this clause may render the bid liable for rejection at the sole discretion of the Employer, and bidders are advised to seek clarification well in advance of the bid due date if any ambiguity is perceived in the requirements of this clause, through the process described under ITB-7 (Clarification of Bidding Documents).

ITB-5: Site Visit

Bidders are advised, at their own cost and risk, to visit and examine the project site and its surroundings, and to obtain all information that may be necessary for preparing the bid and entering into the Contract. A site visit shall be organized on the date specified in the Tender Data Sheet; attendance is strongly recommended but not mandatory.

Bidders should note that non-compliance with this clause may render the bid liable for rejection at the sole discretion of the Employer, and bidders are advised to seek clarification well in advance of the bid due date if any ambiguity is perceived in the requirements of this clause, through the process described under ITB-7 (Clarification of Bidding Documents).

ITB-6: Content of Bidding Documents

The Bidding Documents comprise this Tender Document (Parts A through L), any Addenda issued prior to the bid due date, and the Bidder's own submitted bid documents once accepted. The bidder is expected to examine all instructions, forms, terms and specifications in the Bidding Documents.

Bidders should note that non-compliance with this clause may render the bid liable for rejection at the sole discretion of the Employer, and bidders are advised to seek clarification well in advance of the bid due date if any ambiguity is perceived in the requirements of this clause, through the process described under ITB-7 (Clarification of Bidding Documents).

ITB-7: Clarification of Bidding Documents

A prospective bidder requiring clarification of the Bidding Documents may notify the Employer in writing at the address specified, up to 10 days prior to the bid due date. The Employer shall respond in writing to any request received before this deadline, and shall forward a copy of its response (including an explanation of the query without identifying its source) to all prospective bidders who have received the Bidding Documents.

Bidders should note that non-compliance with this clause may render the bid liable for rejection at the sole discretion of the Employer, and bidders are advised to seek clarification well in advance of the bid due date if any ambiguity is perceived in the requirements of this clause, through the process described under ITB-7 (Clarification of Bidding Documents).

ITB-8: Amendment of Bidding Documents

At any time prior to the bid due date, the Employer may, for any reason, whether at its own initiative or in response to a clarification requested by a prospective bidder, amend the Bidding Documents by issuing an Addendum, which shall be binding on all bidders and shall be published on the e-Procurement portal.

Bidders should note that non-compliance with this clause may render the bid liable for rejection at the sole discretion of the Employer, and bidders are advised to seek clarification well in advance of the bid due date if any ambiguity is perceived in the requirements of this clause, through the process described under ITB-7 (Clarification of Bidding Documents).

ITB-9: Language of Bid

The bid, and all correspondence and documents relating to the bid exchanged between the bidder and the Employer, shall be written in the English language. Supporting documents and printed literature furnished by the bidder may be in another language, provided they are accompanied by an accurate English translation, in which case the English translation shall govern for purposes of interpretation.

Bidders should note that non-compliance with this clause may render the bid liable for rejection at the sole discretion of the Employer, and bidders are advised to seek clarification well in advance of the bid due date if any ambiguity is perceived in the requirements of this clause, through the process described under ITB-7 (Clarification of Bidding Documents).

ITB-10: Documents Comprising the Bid

The bid shall comprise a Technical Bid (Envelope 1) and a Price Bid (Envelope 2), each prepared in accordance with the format and content requirements set out in Part K (Standard Forms) of this Tender Document.

Bidders should note that non-compliance with this clause may render the bid liable for rejection at the sole discretion of the Employer, and bidders are advised to seek clarification well in advance of the bid due date if any ambiguity is perceived in the requirements of this clause, through the process described under ITB-7 (Clarification of Bidding Documents).

ITB-11: Bid Prices

The Contract shall be for the whole scope of work as described in Part D, on a Design-Build lump-sum basis for the capital works, and an annual fixed-plus-variable fee basis for the 10-year O&M period, as further detailed in Part I (BOQ / Price Schedule).

Bidders should note that non-compliance with this clause may render the bid liable for rejection at the sole discretion of the Employer, and bidders are advised to seek clarification well in advance of the bid due date if any ambiguity is perceived in the requirements of this clause, through the process described under ITB-7 (Clarification of Bidding Documents).

ITB-12: Currencies of Bid and Payment

The bidder shall quote its prices entirely in Indian Rupees (INR). Payment under the Contract shall be made in Indian Rupees.

Bidders should note that non-compliance with this clause may render the bid liable for rejection at the sole discretion of the Employer, and bidders are advised to seek clarification well in advance of the bid due date if any ambiguity is perceived in the requirements of this clause, through the process described under ITB-7 (Clarification of Bidding Documents).

ITB-13: Bid Security (EMD)

The bidder shall furnish, as part of its bid, a Bid Security (Earnest Money Deposit) in the amount and form specified in Part F. The Bid Security of unsuccessful bidders shall be returned within 30 days of the expiry of the bid validity period.

Bidders should note that non-compliance with this clause may render the bid liable for rejection at the sole discretion of the Employer, and bidders are advised to seek clarification well in advance of the bid due date if any ambiguity is perceived in the requirements of this clause, through the process described under ITB-7 (Clarification of Bidding Documents).

ITB-14: Period of Validity of Bids

Bids shall remain valid for the period specified in Part F from the bid due date. A bid valid for a shorter period shall be rejected by the Employer as non-responsive.

Bidders should note that non-compliance with this clause may render the bid liable for rejection at the sole discretion of the Employer, and bidders are advised to seek clarification well in advance of the bid due date if any ambiguity is perceived in the requirements of this clause, through the process described under ITB-7 (Clarification of Bidding Documents).

ITB-15: Format and Signing of Bid

The bidder shall prepare one original and one copy of the Technical Bid and the Price Bid, clearly marking each "ORIGINAL" or "COPY" as appropriate, and shall submit the bid electronically through the e-Procurement portal with digital signature of the authorized signatory on all documents.

Bidders should note that non-compliance with this clause may render the bid liable for rejection at the sole discretion of the Employer, and bidders are advised to seek clarification well in advance of the bid due date if any ambiguity is perceived in the requirements of this clause, through the process described under ITB-7 (Clarification of Bidding Documents).

ITB-16: Sealing and Marking of Bids

As the bid is to be submitted electronically, the Technical Bid and Price Bid shall be uploaded as separate, distinctly named and digitally sealed electronic files/envelopes on the e-Procurement portal, in accordance with the portal's user guidelines.

Bidders should note that non-compliance with this clause may render the bid liable for rejection at the sole discretion of the Employer, and bidders are advised to seek clarification well in advance of the bid due date if any ambiguity is perceived in the requirements of this clause, through the process described under ITB-7 (Clarification of Bidding Documents).

ITB-17: Deadline for Submission of Bids

Bids must be received by the Employer at the electronic address specified, no later than the date and time stated in Part A of the Tender Data Sheet. The Employer may, at its discretion, extend this deadline by issuing an Addendum, in which case all rights and obligations of the Employer and the bidders will thereafter be subject to the deadline as extended.

Bidders should note that non-compliance with this clause may render the bid liable for rejection at the sole discretion of the Employer, and bidders are advised to seek clarification well in advance of the bid due date if any ambiguity is perceived in the requirements of this clause, through the process described under ITB-7 (Clarification of Bidding Documents).

ITB-18: Late Bids

Any bid received by the Employer's e-Procurement portal after the deadline for submission of bids prescribed shall be rejected and shall not be opened, as the portal automatically disables submission after the stipulated deadline.

Bidders should note that non-compliance with this clause may render the bid liable for rejection at the sole discretion of the Employer, and bidders are advised to seek clarification well in advance of the bid due date if any ambiguity is perceived in the requirements of this clause, through the process described under ITB-7 (Clarification of Bidding Documents).

ITB-19: Bid Opening

The Employer shall open the Technical Bids in the presence of bidders' representatives who choose to attend (virtually or in person) at the date, time and location specified. Price Bids shall be opened at a later date, only for bidders who are determined to be technically qualified.

Bidders should note that non-compliance with this clause may render the bid liable for rejection at the sole discretion of the Employer, and bidders are advised to seek clarification well in advance of the bid due date if any ambiguity is perceived in the requirements of this clause, through the process described under ITB-7 (Clarification of Bidding Documents).

ITB-20: Process to be Confidential

Information relating to the examination, evaluation and comparison of bids, and recommendations for the award of Contract, shall not be disclosed to bidders or any other persons not officially concerned with the process until the notification of award is made to all bidders.

Bidders should note that non-compliance with this clause may render the bid liable for rejection at the sole discretion of the Employer, and bidders are advised to seek clarification well in advance of the bid

due date if any ambiguity is perceived in the requirements of this clause, through the process described under ITB-7 (Clarification of Bidding Documents).

ITB-21: Clarification of Bids

To assist in the examination, evaluation and comparison of bids, the Employer may, at its discretion, ask any bidder for clarification of its bid. Any clarification submitted by a bidder in respect of its bid that is not in response to a request by the Employer shall not be considered.

Bidders should note that non-compliance with this clause may render the bid liable for rejection at the sole discretion of the Employer, and bidders are advised to seek clarification well in advance of the bid due date if any ambiguity is perceived in the requirements of this clause, through the process described under ITB-7 (Clarification of Bidding Documents).

ITB-22: Correction of Errors

Arithmetical errors in the Price Bid shall be rectified on the following basis: if there is a discrepancy between the unit price and the total price obtained by multiplying the unit price and quantity, the unit price shall prevail, and the total price shall be corrected; if there is a discrepancy between words and figures, the amount in words shall prevail.

Bidders should note that non-compliance with this clause may render the bid liable for rejection at the sole discretion of the Employer, and bidders are advised to seek clarification well in advance of the bid due date if any ambiguity is perceived in the requirements of this clause, through the process described under ITB-7 (Clarification of Bidding Documents).

ITB-23: Contacting the Employer

No bidder shall contact the Employer on any matter relating to its bid from the time of bid opening to the time the Contract is awarded. Any effort by a bidder to influence the Employer in the Employer's bid evaluation, bid comparison or Contract award decisions may result in the rejection of the bidder's bid.

Bidders should note that non-compliance with this clause may render the bid liable for rejection at the sole discretion of the Employer, and bidders are advised to seek clarification well in advance of the bid due date if any ambiguity is perceived in the requirements of this clause, through the process described under ITB-7 (Clarification of Bidding Documents).

ITB-24: Award Criteria

The Employer shall award the Contract to the bidder whose bid has been determined to be substantially responsive to the Bidding Documents, meets the eligibility and qualification criteria, and offers the lowest evaluated Combined Cost (Capital Cost + Net Present Value of 10-Year O&M Cost) among technically qualified bidders.

Bidders should note that non-compliance with this clause may render the bid liable for rejection at the sole discretion of the Employer, and bidders are advised to seek clarification well in advance of the bid due date if any ambiguity is perceived in the requirements of this clause, through the process described under ITB-7 (Clarification of Bidding Documents).

PART B - General Conditions of Contract (GCC)

The General Conditions set out below shall apply to the Contract resulting from this Tender, save as expressly modified or supplemented by the Special Conditions of Contract (Part C).

GCC-1: Definitions

In the Contract, the following terms shall have the meanings assigned to them, unless the context otherwise requires: "Contract" means the Contract Agreement, these General Conditions, the Special Conditions, the Specifications, the Drawings, the Bill of Quantities, and any other documents listed in the Contract Agreement; "Contract Price" means the price payable to the Contractor as specified in the Contract Agreement, subject to such additions and adjustments as provided in the Contract; "Employer" means the Gujraat Water Supply & Sewerage Board; "Contractor" means the person or entity whose bid to perform the Contract has been accepted by the Employer.

This Clause shall be read together with the corresponding provisions of the Special Conditions of Contract (Part C) and the Contract Data Sheet, which may modify, supplement or particularize the general position stated above for the specific circumstances of this Project. In the event of any conflict between this General Condition and a Special Condition, the Special Condition shall prevail to the extent of the conflict.

GCC-2: Interpretation

The headings of the Clauses of these Conditions do not form part of the Conditions and shall not be used for their interpretation. Words importing persons or parties shall include firms and corporations, and any organization having legal capacity. Words in the singular also include the plural and vice versa where the context requires.

This Clause shall be read together with the corresponding provisions of the Special Conditions of Contract (Part C) and the Contract Data Sheet, which may modify, supplement or particularize the general position stated above for the specific circumstances of this Project. In the event of any conflict between this General Condition and a Special Condition, the Special Condition shall prevail to the extent of the conflict.

GCC-3: Language and Law

The Contract shall be governed by the laws of the Republic of India and interpreted in accordance therewith. The language of the Contract shall be English.

This Clause shall be read together with the corresponding provisions of the Special Conditions of Contract (Part C) and the Contract Data Sheet, which may modify, supplement or particularize the general position stated above for the specific circumstances of this Project. In the event of any conflict between this General Condition and a Special Condition, the Special Condition shall prevail to the extent of the conflict.

GCC-4: Notices

Any notice given by one party to the other pursuant to the Contract shall be in writing to the address specified for that purpose in the Contract Data Sheet. A notice shall be effective when delivered or on the notice's effective date, whichever is later.

This Clause shall be read together with the corresponding provisions of the Special Conditions of Contract (Part C) and the Contract Data Sheet, which may modify, supplement or particularize the general position stated above for the specific circumstances of this Project. In the event of any conflict between this General Condition and a Special Condition, the Special Condition shall prevail to the extent of the conflict.

GCC-5: Governing Language

The Contract has been executed in the English language, which shall be the binding and controlling language for all matters relating to the meaning or interpretation of the Contract.

This Clause shall be read together with the corresponding provisions of the Special Conditions of Contract (Part C) and the Contract Data Sheet, which may modify, supplement or particularize the general position stated above for the specific circumstances of this Project. In the event of any conflict between this General Condition and a Special Condition, the Special Condition shall prevail to the extent of the conflict.

GCC-6: Assignment

Neither party to the Contract shall assign the Contract or any part thereof, or any benefit or interest thereunder, without the prior written consent of the other party, save and except that the Contractor shall be entitled to assign to a bank or financial institution its right to any moneys due under the Contract, for the purpose of securing project financing.

This Clause shall be read together with the corresponding provisions of the Special Conditions of Contract (Part C) and the Contract Data Sheet, which may modify, supplement or particularize the general position stated above for the specific circumstances of this Project. In the event of any conflict between this General Condition and a Special Condition, the Special Condition shall prevail to the extent of the conflict.

GCC-7: Sub-Contracting

The Contractor may sub-contract portions of the Works with the Employer's prior written consent, but shall not sub-contract the whole of the Works. Any sub-contracting shall not relieve the Contractor from any liability or obligation under the Contract, and the Contractor shall be responsible for the acts, defaults and negligence of any sub-contractor.

This Clause shall be read together with the corresponding provisions of the Special Conditions of Contract (Part C) and the Contract Data Sheet, which may modify, supplement or particularize the general position stated above for the specific circumstances of this Project. In the event of any conflict between this General Condition and a Special Condition, the Special Condition shall prevail to the extent of the conflict.

GCC-8: Standards

The Goods and Services supplied under the Contract shall conform to the standards mentioned in the Technical Specifications (Part E), and, when no applicable standard is mentioned, to the authoritative standards appropriate to the country of origin of the Goods, such as relevant Bureau of Indian Standards (BIS) codes and Central Public Health & Environmental Engineering Organisation (CPHEEO) manual guidelines.

This Clause shall be read together with the corresponding provisions of the Special Conditions of Contract (Part C) and the Contract Data Sheet, which may modify, supplement or particularize the general position stated above for the specific circumstances of this Project. In the event of any conflict between this General Condition and a Special Condition, the Special Condition shall prevail to the extent of the conflict.

GCC-9: Patent Indemnity

The Contractor shall indemnify the Employer against all third-party claims of infringement of patent, trademark or industrial design rights arising from the use of the Goods or any part thereof in India.

This Clause shall be read together with the corresponding provisions of the Special Conditions of Contract (Part C) and the Contract Data Sheet, which may modify, supplement or particularize the general position stated above for the specific circumstances of this Project. In the event of any conflict between this General Condition and a Special Condition, the Special Condition shall prevail to the extent of the conflict.

GCC-10: Performance Security

Within 21 days of the notification of Contract award, the successful bidder shall furnish a Performance Security in the amount and form specified in the Contract Data Sheet, valid until 90 days beyond the expiry of the O&M period, to guarantee its obligations under the Contract.

This Clause shall be read together with the corresponding provisions of the Special Conditions of Contract (Part C) and the Contract Data Sheet, which may modify, supplement or particularize the general position stated above for the specific circumstances of this Project. In the event of any conflict between this General Condition and a Special Condition, the Special Condition shall prevail to the extent of the conflict.

GCC-11: Inspections and Tests

The Employer or its representative shall have the right to inspect and/or test the Goods to confirm their conformity to the Contract specifications, at no extra cost to the Employer, at the point of delivery and/or at the Goods' final destination, and during manufacture where applicable.

This Clause shall be read together with the corresponding provisions of the Special Conditions of Contract (Part C) and the Contract Data Sheet, which may modify, supplement or particularize the general position stated above for the specific circumstances of this Project. In the event of any conflict between this General Condition and a Special Condition, the Special Condition shall prevail to the extent of the conflict.

GCC-12: Packing

The Contractor shall provide such packing of the Goods as is required to prevent their damage or deterioration during transit to their final destination, as indicated in the Contract, and suitable for such extremes of temperature, salt and precipitation as may be expected during transit and open storage.

This Clause shall be read together with the corresponding provisions of the Special Conditions of Contract (Part C) and the Contract Data Sheet, which may modify, supplement or particularize the general position stated above for the specific circumstances of this Project. In the event of any conflict between this General Condition and a Special Condition, the Special Condition shall prevail to the extent of the conflict.

GCC-13: Delivery and Documents

Delivery of the Goods and completion of Services shall be made by the Contractor in accordance with the schedule specified in the Implementation Schedule (Part D). Delivery documents shall be furnished as specified in the Contract Data Sheet.

This Clause shall be read together with the corresponding provisions of the Special Conditions of Contract (Part C) and the Contract Data Sheet, which may modify, supplement or particularize the general position stated above for the specific circumstances of this Project. In the event of any conflict between this General Condition and a Special Condition, the Special Condition shall prevail to the extent of the conflict.

GCC-14: Insurance

The Works, together with materials and equipment for incorporation therein, shall be insured against loss or damage from any cause whatsoever, other than the risks defined as excepted risks, and shall be for the full replacement cost. The Contractor shall also maintain Third Party Liability Insurance and Workmen's Compensation Insurance as specified in the Special Conditions.

This Clause shall be read together with the corresponding provisions of the Special Conditions of Contract (Part C) and the Contract Data Sheet, which may modify, supplement or particularize the general position stated above for the specific circumstances of this Project. In the event of any conflict between this General Condition and a Special Condition, the Special Condition shall prevail to the extent of the conflict.

GCC-15: Transportation

Unless otherwise specified in the Contract, responsibility for arranging transportation of the Goods, including insurance, shall be in accordance with the delivery terms specified in the Contract.

This Clause shall be read together with the corresponding provisions of the Special Conditions of Contract (Part C) and the Contract Data Sheet, which may modify, supplement or particularize the general position stated above for the specific circumstances of this Project. In the event of any conflict between this General Condition and a Special Condition, the Special Condition shall prevail to the extent of the conflict.

GCC-16: Incidental Services

The Contractor shall be required to provide incidental services including furnishing of tools required for assembly and/or maintenance of the Goods; performance/supervision of on-site assembly and/or start-up of the supplied Goods; and training of the Employer's personnel in the operation and/or maintenance of the Goods.

This Clause shall be read together with the corresponding provisions of the Special Conditions of Contract (Part C) and the Contract Data Sheet, which may modify, supplement or particularize the general position stated above for the specific circumstances of this Project. In the event of any conflict between this General Condition and a Special Condition, the Special Condition shall prevail to the extent of the conflict.

GCC-17: Spare Parts

The Contractor shall be required to provide the mandatory spare parts as listed in the technical specifications and offered in the bid, and shall carry an inventory of spare parts sufficient to support the operation of the Works for the duration of the O&M period.

This Clause shall be read together with the corresponding provisions of the Special Conditions of Contract (Part C) and the Contract Data Sheet, which may modify, supplement or particularize the general position stated above for the specific circumstances of this Project. In the event of any conflict between this General Condition and a Special Condition, the Special Condition shall prevail to the extent of the conflict.

GCC-18: Warranty

The Contractor warrants that the Goods supplied under the Contract are new, unused, and incorporate all recent improvements in design and materials, and shall remain free from defects arising from any act or omission of the Contractor for a Defect Liability Period as specified in the Special Conditions.

This Clause shall be read together with the corresponding provisions of the Special Conditions of Contract (Part C) and the Contract Data Sheet, which may modify, supplement or particularize the general position stated above for the specific circumstances of this Project. In the event of any conflict between this General Condition and a Special Condition, the Special Condition shall prevail to the extent of the conflict.

GCC-19: Payment

Payments shall be made in the currency specified in the Contract, and in the manner specified in the Special Conditions and the Milestone Payment Schedule, on submission of the required documentation and certification of the relevant milestone or O&M period by the Employer's Engineer-in-Charge.

This Clause shall be read together with the corresponding provisions of the Special Conditions of Contract (Part C) and the Contract Data Sheet, which may modify, supplement or particularize the general position stated above for the specific circumstances of this Project. In the event of any conflict between this General Condition and a Special Condition, the Special Condition shall prevail to the extent of the conflict.

GCC-20: Prices

Prices charged by the Contractor for Goods delivered and Services performed under the Contract shall not, with the exception of any price adjustment authorized in the Special Conditions, vary from the prices quoted by the Contractor in its bid.

This Clause shall be read together with the corresponding provisions of the Special Conditions of Contract (Part C) and the Contract Data Sheet, which may modify, supplement or particularize the general position stated above for the specific circumstances of this Project. In the event of any conflict between this General Condition and a Special Condition, the Special Condition shall prevail to the extent of the conflict.

GCC-21: Change Orders

The Employer may at any time, by written order, make changes within the general scope of the Contract in any one or more of the following: drawings, designs or specifications; method of shipment or packing; place of delivery; or the services to be provided. If any such change causes an increase or decrease in the cost of, or the time required for, the Contractor's performance, an equitable adjustment shall be made.

This Clause shall be read together with the corresponding provisions of the Special Conditions of Contract (Part C) and the Contract Data Sheet, which may modify, supplement or particularize the general position stated above for the specific circumstances of this Project. In the event of any conflict between this General Condition and a Special Condition, the Special Condition shall prevail to the extent of the conflict.

GCC-22: Contract Amendments

Subject to GCC-21, no variation in or modification of the terms of the Contract shall be made except by written amendment signed by the parties.

This Clause shall be read together with the corresponding provisions of the Special Conditions of Contract (Part C) and the Contract Data Sheet, which may modify, supplement or particularize the general position stated above for the specific circumstances of this Project. In the event of any conflict between this General Condition and a Special Condition, the Special Condition shall prevail to the extent of the conflict.

GCC-23: Extensions of Time

If at any time during the performance of the Contract the Contractor should encounter conditions impeding timely delivery of the Goods or completion of Services, the Contractor shall promptly notify the Employer in writing of the delay, its likely duration, and its cause, and the Employer shall, as soon as practicable after receipt of such notice, evaluate the situation and may at its discretion extend the Contractor's time for performance.

This Clause shall be read together with the corresponding provisions of the Special Conditions of Contract (Part C) and the Contract Data Sheet, which may modify, supplement or particularize the general position stated above for the specific circumstances of this Project. In the event of any conflict

between this General Condition and a Special Condition, the Special Condition shall prevail to the extent of the conflict.

GCC-24: Liquidated Damages

If the Contractor fails to deliver any or all of the Goods or perform the Services within the period(s) specified in the Contract, the Employer shall, without prejudice to its other remedies, deduct from the Contract Price liquidated damages at the rate specified in the Special Conditions, up to a maximum deduction as specified.

This Clause shall be read together with the corresponding provisions of the Special Conditions of Contract (Part C) and the Contract Data Sheet, which may modify, supplement or particularize the general position stated above for the specific circumstances of this Project. In the event of any conflict between this General Condition and a Special Condition, the Special Condition shall prevail to the extent of the conflict.

GCC-25: Termination for Default

The Employer may, without prejudice to any other remedy, terminate the Contract in whole or in part by written notice of default to the Contractor, if the Contractor fails to deliver any or all of the Goods within the period specified, or fails to perform any other obligation under the Contract.

This Clause shall be read together with the corresponding provisions of the Special Conditions of Contract (Part C) and the Contract Data Sheet, which may modify, supplement or particularize the general position stated above for the specific circumstances of this Project. In the event of any conflict between this General Condition and a Special Condition, the Special Condition shall prevail to the extent of the conflict.

GCC-26: Termination for Insolvency

The Employer may at any time terminate the Contract by giving written notice to the Contractor if the Contractor becomes bankrupt or otherwise insolvent, provided that such termination will not prejudice or affect any right of action or remedy which has accrued or will accrue thereafter to the Employer.

This Clause shall be read together with the corresponding provisions of the Special Conditions of Contract (Part C) and the Contract Data Sheet, which may modify, supplement or particularize the general position stated above for the specific circumstances of this Project. In the event of any conflict between this General Condition and a Special Condition, the Special Condition shall prevail to the extent of the conflict.

GCC-27: Termination for Convenience

The Employer may, by written notice sent to the Contractor, terminate the Contract, in whole or in part, at any time for its own convenience. The notice of termination shall specify that termination is for the Employer's convenience, the extent to which performance of the Contractor under the Contract is terminated, and the date upon which such termination becomes effective.

This Clause shall be read together with the corresponding provisions of the Special Conditions of Contract (Part C) and the Contract Data Sheet, which may modify, supplement or particularize the general position stated above for the specific circumstances of this Project. In the event of any conflict between this General Condition and a Special Condition, the Special Condition shall prevail to the extent of the conflict.

GCC-28: Force Majeure

Notwithstanding the provisions of GCC-24, GCC-25 and GCC-26, the Contractor shall not be liable for forfeiture of its Performance Security, liquidated damages, or termination for default if and to the extent that its delay in performance or other failure to perform its obligations under the Contract is the result of an event of Force Majeure, as defined in the Special Conditions.

This Clause shall be read together with the corresponding provisions of the Special Conditions of Contract (Part C) and the Contract Data Sheet, which may modify, supplement or particularize the general position stated above for the specific circumstances of this Project. In the event of any conflict between this General Condition and a Special Condition, the Special Condition shall prevail to the extent of the conflict.

GCC-29: Settlement of Disputes

The Employer and the Contractor shall make every effort to resolve amicably by direct informal negotiation any disagreement or dispute arising between them under or in connection with the Contract. If, after 30 days, the parties have failed to resolve their dispute amicably, either party may refer the matter to arbitration in accordance with the provisions specified in the Special Conditions.

This Clause shall be read together with the corresponding provisions of the Special Conditions of Contract (Part C) and the Contract Data Sheet, which may modify, supplement or particularize the general position stated above for the specific circumstances of this Project. In the event of any conflict between this General Condition and a Special Condition, the Special Condition shall prevail to the extent of the conflict.

GCC-30: Limitation of Liability

Except in cases of criminal negligence or willful misconduct, and except in the case of any provisions of the Contract related to indemnification, in no event shall either party be liable to the other party for any indirect or consequential loss or damage, loss of use, loss of production, or loss of profits, related to or arising out of the Contract.

This Clause shall be read together with the corresponding provisions of the Special Conditions of Contract (Part C) and the Contract Data Sheet, which may modify, supplement or particularize the general position stated above for the specific circumstances of this Project. In the event of any conflict between this General Condition and a Special Condition, the Special Condition shall prevail to the extent of the conflict.

PART C - Special Conditions of Contract (SCC)

The following Special Conditions shall supplement and, where inconsistent, take precedence over the General Conditions of Contract (Part B) for this specific Project.

SCC-1: Employer & Engineer-in-Charge

The Employer is the Gujraat Water Supply & Sewerage Board (GWSSB). The Engineer-in-Charge for this Contract shall be the Superintending Engineer, Surat Circle, GWSSB, or such other officer as may be notified in writing from time to time.

Compliance with this Special Condition will be specifically assessed by the Employer's Engineer-in-Charge during the relevant phase of the Contract (Design-Build or Operation & Maintenance, as applicable), and any deviation proposed by the Contractor at the bid stage against this Clause shall be explicitly and separately declared in the Contractor's Deviation Statement, failing which the Contractor shall be deemed to have accepted this Clause in full and without qualification.

SCC-2: Site Description

The Site for the Water Treatment Plant is located at Village Kosad, Surat, on land measuring approximately 18 hectares, already in possession of the Employer and to be handed over to the Contractor within 15 days of the Notice to Proceed, free from encumbrances, save as may be described in the Site Data Report (Annexure).

Compliance with this Special Condition will be specifically assessed by the Employer's Engineer-in-Charge during the relevant phase of the Contract (Design-Build or Operation & Maintenance, as applicable), and any deviation proposed by the Contractor at the bid stage against this Clause shall be explicitly and separately declared in the Contractor's Deviation Statement, failing which the Contractor shall be deemed to have accepted this Clause in full and without qualification.

SCC-3: Design Life

The Works shall be designed for a minimum design life of 30 years for civil structures and process units, and 15 years for mechanical/electrical equipment (with major overhaul provisioned at Year 15), in accordance with CPHEEO Manual on Water Supply and Treatment guidelines.

Compliance with this Special Condition will be specifically assessed by the Employer's Engineer-in-Charge during the relevant phase of the Contract (Design-Build or Operation & Maintenance, as applicable), and any deviation proposed by the Contractor at the bid stage against this Clause shall be explicitly and separately declared in the Contractor's Deviation Statement, failing which the Contractor shall be deemed to have accepted this Clause in full and without qualification.

SCC-4: Raw Water Source

Raw water shall be drawn from the Tapi River at the existing GWSSB intake well located approximately 2.8 km from the WTP site; raw water quality data for the preceding 3 years is provided in Annexure - Raw Water Characteristics.

Compliance with this Special Condition will be specifically assessed by the Employer's Engineer-in-Charge during the relevant phase of the Contract (Design-Build or Operation & Maintenance, as applicable), and any deviation proposed by the Contractor at the bid stage against this Clause shall be explicitly and separately declared in the Contractor's Deviation Statement, failing which the Contractor shall be deemed to have accepted this Clause in full and without qualification.

SCC-5: Treated Water Quality Standards

Treated water quality shall conform to IS 10500:2012 (Drinking Water Specification) at all times, with the specific performance parameters and guarantee values set out in Part H (O&M Requirements) of this Tender Document.

Compliance with this Special Condition will be specifically assessed by the Employer's Engineer-in-Charge during the relevant phase of the Contract (Design-Build or Operation & Maintenance, as applicable), and any deviation proposed by the Contractor at the bid stage against this Clause shall be explicitly and separately declared in the Contractor's Deviation Statement, failing which the Contractor shall be deemed to have accepted this Clause in full and without qualification.

SCC-6: Performance Security Amount

The Performance Security shall be 5% of the Design-Build Contract Price plus 5% of the total 10-year O&M Contract Price (Net Present Value), to be furnished as separate Bank Guarantees for the D&B phase and the O&M phase respectively.

Compliance with this Special Condition will be specifically assessed by the Employer's Engineer-in-Charge during the relevant phase of the Contract (Design-Build or Operation & Maintenance, as applicable), and any deviation proposed by the Contractor at the bid stage against this Clause shall be explicitly and separately declared in the Contractor's Deviation Statement, failing which the Contractor shall be deemed to have accepted this Clause in full and without qualification.

SCC-7: Defect Liability Period

The Defect Liability Period for civil works shall be 24 months, and for mechanical/electrical equipment shall be 18 months, both commencing from the date of Provisional Acceptance, running concurrently with the commencement of the O&M period.

Compliance with this Special Condition will be specifically assessed by the Employer's Engineer-in-Charge during the relevant phase of the Contract (Design-Build or Operation & Maintenance, as applicable), and any deviation proposed by the Contractor at the bid stage against this Clause shall be explicitly and separately declared in the Contractor's Deviation Statement, failing which the Contractor shall be deemed to have accepted this Clause in full and without qualification.

SCC-8: Liquidated Damages Rate

Liquidated damages for delay in completion of the Design-Build phase shall be levied at 0.5% of the D&B Contract Price per week of delay, subject to a maximum of 10% of the D&B Contract Price. Liquidated damages for O&M performance shortfalls shall be as set out in Part H.

Compliance with this Special Condition will be specifically assessed by the Employer's Engineer-in-Charge during the relevant phase of the Contract (Design-Build or Operation & Maintenance, as applicable), and any deviation proposed by the Contractor at the bid stage against this Clause shall be explicitly and separately declared in the Contractor's Deviation Statement, failing which the Contractor shall be deemed to have accepted this Clause in full and without qualification.

SCC-9: Price Adjustment during O&M

The O&M annual fee shall be subject to an annual price escalation linked to the Wholesale Price Index (WPI) for the preceding financial year, capped at a maximum of 7% per annum, applied from the second year of the O&M period onward.

Compliance with this Special Condition will be specifically assessed by the Employer's Engineer-in-Charge during the relevant phase of the Contract (Design-Build or Operation & Maintenance, as applicable), and any deviation proposed by the Contractor at the bid stage against this Clause shall be explicitly and separately declared in the Contractor's Deviation Statement, failing which the Contractor shall be deemed to have accepted this Clause in full and without qualification.

SCC-10: Power Supply During O&M

Electrical power for plant operation during the O&M period shall be arranged and paid for by the Employer; the Contractor shall be responsible for the efficient use of power and shall report any abnormal consumption to the Employer.

Compliance with this Special Condition will be specifically assessed by the Employer's Engineer-in-Charge during the relevant phase of the Contract (Design-Build or Operation & Maintenance, as applicable), and any deviation proposed by the Contractor at the bid stage against this Clause shall be explicitly and separately declared in the Contractor's Deviation Statement, failing which the Contractor shall be deemed to have accepted this Clause in full and without qualification.

SCC-11: Chemicals & Consumables

The Contractor shall be responsible for procurement, storage, and use of all chemicals (alum/PAC, lime, chlorine, polyelectrolyte) required for plant operation during the O&M period, at its own cost, factored into the quoted O&M fee, subject to the specific consumption norms specified in Part H.

Compliance with this Special Condition will be specifically assessed by the Employer's Engineer-in-Charge during the relevant phase of the Contract (Design-Build or Operation & Maintenance, as applicable), and any deviation proposed by the Contractor at the bid stage against this Clause shall be explicitly and separately declared in the Contractor's Deviation Statement, failing which the Contractor shall be deemed to have accepted this Clause in full and without qualification.

SCC-12: Sludge Disposal

The Contractor shall be responsible for dewatering and disposal of water treatment sludge in an environmentally compliant manner, in accordance with Gujraat Pollution Control Board (GPCB) norms, at its own cost during the O&M period.

Compliance with this Special Condition will be specifically assessed by the Employer's Engineer-in-Charge during the relevant phase of the Contract (Design-Build or Operation & Maintenance, as applicable), and any deviation proposed by the Contractor at the bid stage against this Clause shall be explicitly and separately declared in the Contractor's Deviation Statement, failing which the Contractor shall be deemed to have accepted this Clause in full and without qualification.

SCC-13: Performance Guarantee Tests

Upon completion of the Design-Build phase, a 30-day Performance Guarantee Test Run shall be conducted jointly by the Contractor and the Employer, during which the plant shall demonstrate the guaranteed capacity, treated water quality, and specific power/chemical consumption parameters specified in Part H.

Compliance with this Special Condition will be specifically assessed by the Employer's Engineer-in-Charge during the relevant phase of the Contract (Design-Build or Operation & Maintenance, as applicable), and any deviation proposed by the Contractor at the bid stage against this Clause shall be explicitly and separately declared in the Contractor's Deviation Statement, failing which the Contractor shall be deemed to have accepted this Clause in full and without qualification.

SCC-14: Handback Condition at End of O&M

At the end of the 10-year O&M period, the Contractor shall hand back the Works to the Employer in good operating condition, with a residual design life of not less than 5 years for major mechanical/electrical equipment, subject to a joint condition assessment to be carried out in the final year of the O&M period.

Compliance with this Special Condition will be specifically assessed by the Employer's Engineer-in-Charge during the relevant phase of the Contract (Design-Build or Operation & Maintenance, as applicable), and any deviation proposed by the Contractor at the bid stage against this Clause shall be explicitly and separately declared in the Contractor's Deviation Statement, failing which the Contractor shall be deemed to have accepted this Clause in full and without qualification.

SCC-15: Integrity Pact

This Contract is subject to an Integrity Pact between the Employer and the Contractor, in the format prescribed in Part K, aimed at preventing corrupt practices in the procurement process.

Compliance with this Special Condition will be specifically assessed by the Employer's Engineer-in-Charge during the relevant phase of the Contract (Design-Build or Operation & Maintenance, as applicable), and any deviation proposed by the Contractor at the bid stage against this Clause shall be explicitly and separately declared in the Contractor's Deviation Statement, failing which the Contractor shall be deemed to have accepted this Clause in full and without qualification.

PART D - Scope of Work

The Contractor shall carry out the complete Design-Build of a 100 MLD Conventional Water Treatment Plant at Kosad, Surat, together with associated raw water pumping station, clear water pumping station, 8.5 km rising main to the city distribution network, SCADA-based automation, and shall subsequently operate and maintain the completed Works for a period of 10 years, covering the following broad scope, described in further detail in the subsections below:

D.1 Detailed Engineering

Detailed engineering, hydraulic design, and process design of all treatment units. The Contractor shall carry out complete detailed engineering including hydraulic profile calculation across the entire treatment train, process design of each unit operation, structural design of all civil structures, and preparation of Good-for-Construction (GFC) drawings for the Employer's review and approval prior to commencement of any construction activity.

D.2 Raw Water Pumping Station

Intake works, pumps, rising main to WTP. This includes tie-in works at the existing GWSSB intake well, a new raw water pumping station with the pump configuration specified in Part E.2, and a raw water rising main conveying water to the WTP inlet works, complete with all civil, mechanical, electrical and instrumentation works.

D.3 Pre-treatment

Screening, aeration (if required based on raw water quality). Coarse and fine screening shall be provided at the WTP inlet to remove floating debris; aeration shall be provided if deemed necessary based on the Contractor's own assessment of raw water quality (particularly during low-flow, high-organic-load periods), subject to Employer's design review.

D.4 Coagulation-Flocculation

Rapid mix + flocculation basins (clariflocculators). Coagulant (alum or poly-aluminium chloride) shall be dosed at a rapid mix chamber, followed by flocculation in the flocculation zone of the clariflocculator units, designed per the criteria in Part E.3 and Annexure L-7 (CPHEEO extract).

D.5 Sedimentation / Clarification

Clariflocculator units with tube settlers. Clarification shall be achieved in the same clariflocculator units (combined flocculation-clarification design), fitted with tube settler modules to enhance settling efficiency and reduce the footprint required, per the specification in Part E.3.

D.6 Filtration

Rapid Gravity Sand Filters (RGF) with automatic backwashing. Filtration shall be through dual-media Rapid Gravity Filters as specified in Part E.4, with fully automatic, PLC-controlled sequential backwashing to minimize manual intervention and ensure consistent filtered water quality.

D.7 Disinfection

Chlorination system (gas chlorination with scrubber, or sodium hypochlorite as alternative). The primary disinfection system shall be gas chlorination as specified in Part E.5, with all associated safety systems; bidders proposing sodium hypochlorite dosing as an alternative shall clearly declare this as a deviation and justify the equivalent disinfection efficacy and cost implications.

D.8 Sludge Handling

Sludge thickening, mechanical dewatering (centrifuge/filter press), sludge drying beds. Sludge generated from the clariflocculators shall be thickened, mechanically dewatered to a minimum cake dryness of 20%, with sludge drying beds provided as a backup/overflow system during peak monsoon sludge loads, per Part E.6.

D.9 Clear Water Reservoir (CWR)

Minimum 4-hour storage capacity at design flow. The CWR shall provide a minimum of 4 hours of balancing storage at the design flow of 100 MLD, constructed as an RCC underground/semi-underground water-retaining structure per the civil specifications in Part L-A.

D.10 Clear Water Pumping Station

Pumps to convey treated water via the rising main to the city distribution system. As specified in Part E.7, the clear water pumping station shall be designed with adequate redundancy to ensure uninterrupted supply to the city distribution network even during single-pump maintenance/failure scenarios.

D.11 Rising Main

Approximately 8.5 km, MS/DI pipeline as per hydraulic design. The Contractor shall carry out detailed hydraulic transient (surge) analysis and finalize the rising main alignment, pipe material, diameter and class, subject to the indicative parameters in Part E.7 and the Employer's approval of the final alignment (which may traverse public roads requiring additional permissions per Part L-A statutory clearances).

D.12 Electrical Works

HT/LT substation, DG backup, all plant electricals. Complete electrical works from the HT incoming point through to the last LT distribution point shall be provided, per the specifications in Part L-B (M&E Standard Specifications) and Part E.8.

D.13 SCADA/Instrumentation

Complete process automation and remote monitoring. A fully integrated SCADA system covering all process areas, with the architecture and instrumentation requirements specified in Part E.8, shall be provided, commissioned, and demonstrated to meet the specified 99.5% uptime guarantee prior to Provisional Acceptance.

D.14 Laboratory

Fully equipped water quality testing laboratory at the WTP site. A laboratory equipped to test all parameters required under IS 10500:2012, including physical, chemical, and bacteriological testing

capability, shall be provided and staffed as per Part H.1 (O&M Manpower Requirements) throughout the O&M period.

D.15 Ancillary Buildings & Site Works

Administrative building, staff quarters (as applicable), boundary wall, internal roads, landscaping. Ancillary civil works shall be provided as detailed in Part L-A, sized to comfortably accommodate the O&M staffing plan set out in Part H.1, with adequate parking, visitor facilities, and site security infrastructure.

D.16 Operation & Maintenance (10 Years)

Comprehensive O&M including manpower, chemicals, power liaison, and performance guarantees as per Part H. Following Provisional Acceptance, the Contractor shall operate and maintain the complete Works for 10 years, meeting the performance guarantees, manpower norms, and reporting requirements detailed in Part H, and shall hand back the Works to the Employer at the end of the O&M period in the condition specified in SCC-14.

PART E - Technical Specifications

E.1 Design Basis Parameters

Parameter	Design Value / Requirement
Design Capacity	100 MLD (Million Litres per Day)
Design Horizon Year	2056 (30-year design life)
Raw Water Source	Tapi River, via existing intake well
Raw Water Turbidity Range (design)	10 - 1500 NTU (monsoon peak)
Treated Water Turbidity	≤ 1 NTU (guaranteed ≤ 0.5 NTU, 95th percentile)
Treated Water Residual Chlorine	0.2 - 0.5 mg/l at CWR outlet
Plant Hydraulic Loss (max)	≤ 4.5 m through complete treatment train
Filtration Rate (RGF)	5 - 6 m ³ /m ² /hr at design flow
Clariflocculator Surface Loading	≤ 1.5 m ³ /m ² /hr
Detention Time - Flocculation	30 minutes (minimum)
Detention Time - Clarification	2.5 - 3 hours
Clear Water Reservoir Capacity	16.7 ML minimum (4 hours at design flow)
Power Reliability	Dual HT feeder + 100% DG backup for critical loads
SCADA Availability	≥ 99.5% uptime guaranteed during O&M

E.2 Raw Water Pumping Station - Specification

Parameter	Requirement
Number of Pumps	4 Working + 2 Standby (vertical turbine type)
Individual Pump Capacity	625 lps @ 30m head (each)
Pump Efficiency	≥ 78% at duty point
Motor Rating	As per pump duty, IE3 premium efficiency
Variable Frequency Drives (VFD)	Provided for 2 of 4 working pumps for flow modulation
Screening	Mechanically raked bar screens, 2 working + 1 standby

E.3 Clariflocculator - Specification

Parameter	Requirement
Number of Units	4 (each rated for 25 MLD)
Type	Circular, mechanically agitated, with tube settler modules
Sludge Scraping	Motorized central drive with sludge scraper mechanism
Sludge Withdrawal	Automated, timer/turbidity-based hydraulic sludge withdrawal
Material of Construction (mechanism)	SS 304/316 for wetted parts, MS-FRP coated for structural

E.4 Rapid Gravity Filters - Specification

Parameter	Requirement
Number of Filter Beds	16 Nos. (each approx. 45 sqm filtration area)
Filter Media	Dual media - sand + anthracite, per CPHEEO Manual
Backwash System	Air scour + water wash, automatic sequential backwashing
Backwash Water Source	Dedicated backwash water pumps + elevated backwash tank
Filter Control	PLC-based automatic filter control with SCADA integration
Underdrain System	Nozzle-type underdrain, HDPE/PP nozzles

E.5 Disinfection System - Specification

Parameter	Requirement
Primary Disinfectant	Chlorine gas (Cl ₂) with automatic vacuum feed chlorinators
Chlorine Storage	Ton container storage, minimum 15 days consumption
Chlorine Leak Safety	Gas scrubber system, gas leak detectors, emergency kit as per CPCB norms
Dosing Points	Pre-chlorination (optional) and post-filtration (primary)

Parameter	Requirement
Residual Monitoring	Online chlorine analyzers at CWR inlet and distribution points

E.6 Sludge Handling System - Specification

Parameter	Requirement
Sludge Thickening	Gravity thickener, 2 Nos.
Sludge Dewatering	Centrifuge or Filter Press (bidder's choice), minimum 20% cake dryness
Sludge Drying Beds	Provided as backup, sized for monsoon peak sludge load
Sludge Disposal	As per GPCB norms; Contractor responsible during O&M period

E.7 Clear Water Pumping Station & Rising Main - Specification

Parameter	Requirement
Number of Pumps	5 Working + 2 Standby (horizontal split-case centrifugal)
Individual Pump Capacity	500 lps @ 45m head (each)
Rising Main Length	Approximately 8.5 km
Rising Main Material	MS (welded) / Ductile Iron, as per hydraulic design and soil conditions
Rising Main Diameter	1400mm (indicative, subject to detailed hydraulic design)
Surge Protection	Surge vessels / surge relief valves as per transient analysis

E.8 SCADA & Instrumentation - Specification

Parameter	Requirement
Architecture	PLC-based distributed control with central SCADA server
Instrumentation	Flow meters, level transmitters, turbidity/chlorine/pH analyzers at all critical points
Communication	Fiber optic backbone between process areas and control room
Remote Monitoring	Integration with GWSSB's central monitoring cell (as applicable)
Data Historian	Minimum 5-year data retention for all process parameters
Redundancy	Dual redundant SCADA server, UPS-backed control systems

E.9 Laboratory - Equipment List

The on-site laboratory shall be equipped, at a minimum, with the following instruments and apparatus, to enable full in-house testing of all parameters required under IS 10500:2012 on a routine basis, with only specialized/infrequent parameters (e.g., specific trace organics) sent to an external NABL-accredited laboratory.

Equipment	Purpose
Turbidimeter (Nephelometric)	Turbidity measurement (raw & treated water)
pH Meter	pH measurement
Digital Colorimeter/Spectrophotometer	Colour, iron, and other colorimetric parameters
Jar Test Apparatus	Coagulant dose optimization studies
Total Dissolved Solids (TDS) Meter	TDS/conductivity measurement
Alkalinity/Hardness Titration Kit	Alkalinity and hardness determination
Residual Chlorine Test Kit (DPD method)	Free & total residual chlorine
BOD Incubator	Biochemical Oxygen Demand testing
COD Digestion & Titration Unit	Chemical Oxygen Demand testing
Microbiological Incubator	Bacteriological (coliform/E.coli) testing
Autoclave	Sterilization of lab glassware/media
Laminar Air Flow Cabinet	Aseptic microbiological sample handling
Analytical Balance	Precision weighing for reagent preparation
Atomic Absorption Spectrophotometer (AAS) or	Heavy metals testing (as required)

Equipment	Purpose
ICP	
Fluoride Ion Selective Electrode Meter	Fluoride testing
Nitrate/Nitrite Test Kit	Nitrogen compound testing
Muffle Furnace	Ash content / solids testing
Hot Air Oven	Drying/sterilization
Membrane Filtration Apparatus	Microbiological filtration testing
Refrigerator (sample storage)	Sample preservation

E.10 Administrative Building - Room Schedule

Room / Facility	Approximate Area (sqm)
Reception & Waiting Area	40
O&M Manager's Office	25
Shift In-charge Room	20
Control Room (SCADA)	60
Conference / Training Room	50
Open Office (Engineers)	80
Store Room (spares)	100
Workshop (mechanical/electrical maintenance)	80
Chemical Store (dry chemicals)	60
Male & Female Toilets (each floor)	30
Pantry / Rest Area	25
Security Cabin	12
First Aid Room	15
Server/IT Room	20
Record/Document Room	20

PART F - Eligibility & Qualification Criteria

S.No.	Criteria	Requirement
EC-1	Average Annual Turnover	Minimum ₹100,00,00,000 (₹100 Crore) in each of the last 3 financial years
EC-2	D&B Experience	Successful completion of at least 2 Water Treatment Plants of 50 MLD or above capacity in the last 10 years, OR 1 WTP of 100 MLD or above
EC-3	O&M Experience	Ongoing or completed O&M contract of at least 1 water/wastewater treatment plant of 50 MLD or above, for a minimum continuous period of 3 years
EC-4	Net Worth	Positive net worth, minimum ₹50 Crore as per latest audited financial statement
EC-5	Quality & Environmental Certifications	Valid ISO 9001:2015, ISO 14001:2015, ISO 45001:2018
EC-6	Statutory Registrations	Valid PAN, GST Registration, EPFO Registration, and relevant Pollution Control Board consents/registrations
EC-7	Blacklisting Declaration	Self-certified, notarized undertaking of no blacklisting/debarment by any Central/State Government, PSU, or Urban Local Body
EC-8	EMD Submission	EMD of ₹2,00,00,000 in the form of Bank Guarantee, valid 210 days

PART G - Evaluation Criteria

Evaluation shall be conducted in three sequential stages: (1) Eligibility Screening, (2) Technical Evaluation out of 100 marks with a minimum qualifying score of 65, and (3) Price Evaluation of technically qualified bidders' Combined Cost (D&B Price + NPV of 10-Year O&M Price, discounted at 8% per annum).

Criteria	Max Marks
Relevant D&B Project Experience	20
O&M Track Record & Performance	20
Process Design & Technical Compliance (Part E)	25
Execution Methodology, Schedule & QA/QC Plan	15
O&M Staffing, Manpower Plan & Key Personnel	10
HSE Plan & Environmental Management	10

G.1 Technical Evaluation Sub-Criteria Breakdown

Each of the six major evaluation criteria above shall be further assessed against the sub-criteria described below, to guide the Evaluation Committee in arriving at a consistent and well-documented technical score for each bidder.

Major Criterion	Sub-Criterion	Max Sub-Marks
D&B Project Experience	Number of qualifying 50 MLD+ WTP projects	8
D&B Project Experience	Complexity/similarity of process technology to this Project	6
D&B Project Experience	Timely completion track record on cited projects	6
O&M Track Record	Duration and continuity of O&M contracts held	8
O&M Track Record	Performance guarantee achievement on cited O&M contracts	8
O&M Track Record	Client satisfaction / reference feedback	4
Process Design & Technical Compliance	Compliance with Part E design parameters (unit-by-unit)	12
Process Design & Technical Compliance	Quality and completeness of preliminary design basis report	8
Process Design & Technical Compliance	Innovation / value engineering proposals (if any)	5
Execution Methodology	Realism and detail of the proposed project schedule	6
Execution Methodology	Quality of QA/QC Plan	5
Execution Methodology	Risk identification and mitigation approach	4
O&M Staffing	Adequacy of proposed staffing against Part H.1 minimums	5
O&M Staffing	Qualification and experience of key personnel proposed	5
HSE Plan	Comprehensiveness of HSE Plan (construction phase)	5
HSE Plan	Comprehensiveness of Environmental Management Plan (O&M phase, incl. sludge disposal)	5

Bidders scoring below the minimum qualifying mark of 65/100 in the aggregate, or below 50% of the maximum marks allotted to any single major criterion (Relevant D&B Project Experience, O&M Track Record, or Process Design & Technical Compliance), shall be deemed technically non-responsive and shall not proceed to Price Bid evaluation, regardless of their aggregate score.

PART H - Operation & Maintenance Requirements (10-Year)

The Contractor shall operate and maintain the Works for a period of 10 years from the date of Provisional Acceptance, achieving and maintaining the performance guarantees set out below throughout the O&M period, failing which liquidated damages / fee deductions shall apply as detailed in the Special Conditions.

Performance Parameter	Guaranteed Value	Penalty for Non-Compliance
Plant Availability	≥ 98% of design capacity availability, annual average	Pro-rata O&M fee deduction for shortfall
Treated Water Turbidity	≤ 1 NTU, 95th percentile monthly	₹5,00,000 per exceedance month
Treated Water Residual Chlorine	0.2 - 0.5 mg/l at CWR outlet	₹2,00,000 per exceedance month
IS 10500:2012 Compliance	100% compliance, monthly third-party testing	₹10,00,000 per non-compliant parameter per month
Specific Power Consumption	≤ 0.55 kWh per kL treated water	Excess power cost borne by Contractor beyond guaranteed value
Specific Chemical Consumption (Alum)	≤ 25 mg/l average annual dose	Excess chemical cost borne by Contractor beyond guaranteed value
Unplanned Shutdown Duration	≤ 24 hours per annum, cumulative	₹1,00,000 per additional hour of unplanned shutdown
Sludge Disposal Compliance	100% compliance with GPCB norms	As per GPCB penalty + contractual deduction of ₹5,00,000 per violation

H.1 O&M Manpower Requirements (Minimum)

Role	Minimum Qualification	Minimum Headcount
O&M Manager	B.E. (Civil/Environmental/Chemical), 10+ yrs WTP O&M	1
Shift In-charge (24x7, 4 shifts)	B.E./Diploma, 5+ yrs	4
Process/Chemist	M.Sc. Chemistry / B.Tech, water quality experience	3
Electrical & Instrumentation Engineer	B.E. Electrical/Instrumentation	2
Mechanical Maintenance Engineer	B.E. Mechanical	2
Plant Operators (24x7)	ITI/Diploma, trained operators	12
Laboratory Technicians	B.Sc./Diploma, NABL-aligned lab experience	3
Safety Officer	Diploma with safety certification	1

H.2 Shift Operations & Emergency Response

The Contractor shall operate the WTP on a continuous 24x7x365 basis throughout the O&M period, organized into 4 shifts to ensure adequate handover overlap and continuous supervisory coverage, with the Shift In-charge for each shift holding full authority to take emergency operational decisions (including plant shutdown, if necessary for safety) subject to immediate notification to the O&M Manager and the Employer's Engineer-in-Charge.

An Emergency Response Plan shall be maintained on-site at all times, covering scenarios including but not limited to: chlorine gas leak, major pipeline burst on the rising main, complete power failure exceeding DG backup capacity, raw water quality excursion beyond design parameters, and structural

distress in any major civil structure. Mock drills for the chlorine leak scenario shall be conducted at least quarterly, with records maintained and made available for the Employer's audit.

The Contractor shall submit Daily Operation Reports, Monthly O&M Performance Reports, and an Annual O&M Review Report to the Employer, in formats to be mutually agreed at Contract signing, covering as a minimum: raw and treated water quality test results, chemical and power consumption, plant availability and downtime log with root-cause analysis, manpower deployment record, spare parts consumption and stock status, and any incidents/near-misses recorded during the reporting period.

H.3 Training & Capacity Building

The Contractor shall conduct an annual training and capacity-building program for its own O&M staff, covering process optimization, safety procedures, and SCADA system operation, and shall additionally make available at least 2 training slots per year, at no cost to the Employer, for GWSSB's own departmental staff who may be nominated to shadow the Contractor's O&M team as part of the Employer's institutional capacity-building objectives under the AMRUT 2.0 program.

PART H-A - Risk Allocation Matrix (DBO Contract)

The following Risk Allocation Matrix sets out, for the Employer's and prospective bidders' common understanding, how key project risks are allocated between the Employer and the Contractor under this DBO Contract structure. This matrix is descriptive of the risk allocation embedded in the GCC/SCC provisions and does not itself create additional contractual obligations beyond those provisions.

Risk Description	Primary Allocation	Rationale / Basis
Design deficiency leading to non-achievement of performance guarantees	Contractor	Design-Build responsibility rests fully with the Contractor under GCC-8 and Part E
Land acquisition / handover delay	Employer	Site already in Employer's possession per SCC-2; any residual delay risk rests with Employer
Raw water quality significantly worse than historical data (Annexure L-2)	Shared	Contractor to design with reasonable contingency; extreme deviations beyond design basis may trigger Change Order under GCC-21
Force Majeure events (flood, earthquake, pandemic)	Shared, per GCC-28	Relief from liquidated damages/termination as defined; cost-sharing per Special Conditions where applicable
Construction cost escalation (materials, labour)	Contractor	Firm-price D&B Contract; no price adjustment except as explicitly provided
O&M cost escalation beyond WPI cap	Contractor	O&M fee escalation capped at 7% p.a. per SCC-9; excess borne by Contractor
Power tariff increase during O&M	Employer	Power cost borne directly by Employer per SCC-10; Contractor responsible only for efficient usage
Statutory/regulatory changes affecting scope or cost	Shared	Material adverse statutory changes may qualify for Change Order treatment under GCC-21
Equipment obsolescence during 10-year O&M period	Contractor	Contractor responsible for maintaining designed performance for full O&M term including any necessary component replacement
Third-party damage to Works (e.g., excavation damage to rising main)	Contractor to mitigate; recoverable from responsible party	Contractor to maintain adequate insurance and pursue recovery; Employer to assist with any required permissions/access
Delay in statutory approvals (CEIG-equivalent, pollution consents)	Shared	Contractor to prepare and submit applications; Employer to support and expedite as project proponent where required
Sludge disposal site unavailability	Employer to facilitate; Contractor to execute	Employer to assist in identifying/approving disposal sites; Contractor responsible for compliant disposal operations
Currency/exchange rate risk (imported components, if any)	Contractor	Minimal import content expected; any residual risk borne by Contractor as prices are quoted in INR
Water demand significantly below/above design projections during O&M	Employer	Employer bears volume risk; O&M fee structure is largely fixed-fee based, not purely volumetric
Key personnel unavailability/attrition (Contractor side)	Contractor	Contractor responsible for maintaining the minimum manpower levels specified in Part H.1 throughout the O&M period

PART H-B - Performance Guarantee Test Protocol (Illustrative Steps)

The 30-day Performance Guarantee Test Run referenced in SCC-13 shall follow, as a minimum, the sequence of steps outlined below, to be elaborated into a detailed Test Protocol Document to be jointly agreed between the Employer and the Contractor at least 60 days prior to the scheduled start of the Test Run.

Step No.	Test Protocol Step
1	Joint pre-test inspection of all civil, mechanical, electrical and instrumentation systems for readiness
2	Calibration verification of all critical instruments (flow meters, turbidity/chlorine analyzers, level transmitters)
3	Confirmation of chemical stock (alum, chlorine, polyelectrolyte) sufficient for the full test duration
4	Confirmation of manpower deployment as per the approved O&M staffing plan (Part H.1)
5	Baseline raw water quality sampling and testing (Day 0)
6	Ramp-up of plant to 50% design capacity, sustained for 48 hours, with continuous monitoring
7	Ramp-up of plant to 75% design capacity, sustained for 48 hours, with continuous monitoring
8	Ramp-up of plant to 100% design capacity, sustained for the balance of the 30-day period
9	Daily composite sampling and third-party laboratory testing of treated water against IS 10500:2012 parameters
10	Continuous monitoring and daily logging of specific power consumption (kWh/kL)
11	Continuous monitoring and daily logging of specific chemical consumption (mg/l dosing rates)
12	Verification of SCADA data accuracy against manual/independent measurements at defined checkpoints
13	Simulated single-pump-failure test at both raw water and clear water pumping stations, to verify standby capacity
14	Simulated grid power failure test, to verify DG backup changeover time and capacity
15	Verification of sludge dewatering system performance against the guaranteed cake dryness parameter
16	Verification of chlorine safety systems (leak detection, scrubber activation) through a controlled functional test
17	Compilation of daily test data into a Performance Guarantee Test Report by the Contractor
18	Joint review of the Test Report by the Employer's Engineer-in-Charge and the Contractor's Project Manager
19	Resolution of any performance shortfalls identified, with a defined cure period if required, prior to final sign-off
20	Issuance of the Provisional Acceptance Certificate upon successful demonstration of all guaranteed parameters

PART L-C - List of Employer-Furnished Items

The following items shall be furnished by the Employer, free of cost, to the Contractor for the purposes of this Contract; all other items required for the execution of the Works shall be arranged by the Contractor at its own cost, whether or not explicitly listed in this Tender Document.

Item	Details
Project Site	18 hectares at Kosad, Surat, handed over per SCC-2
Existing Intake Well Tie-in Point	Access for connection of new raw water pumping main
Historical Raw Water Quality Data	36-month record per Annexure L-2
Preliminary Topographic & Geotechnical Data	Per Annexure L-1, for reference only; Contractor to verify independently
Point of Connection for Treated Water Rising Main	At the designated city distribution network tie-in point
Liaison Support for Statutory Approvals	As Employer/project proponent, per Annexure L-A statutory matrix (equivalent water-sector clauses)
Power Tariff Payment (O&M phase)	Direct payment by Employer to Discom, per SCC-10
Access Roads (existing, up to site boundary)	Existing 12m approach road per Annexure L-1; strengthening cost reimbursable as noted

PART I - Bill of Quantities (BOQ) / Price Schedule Format

Bidders shall quote prices in the format prescribed below, covering the Design-Build (D&B) capital cost and the 10-Year O&M cost (Year-wise), in the Price Bid envelope only. The Engineer's Estimate for reference is indicated (not disclosed to bidders during actual bidding, shown here for BASE_DOC completeness):

Component	Engineer's Estimate (₹)
Civil Works (all structures)	₹1,45,00,00,000
Mechanical Equipment (pumps, clariflocculators, filters, etc.)	₹98,00,00,000
Electrical & SCADA	₹42,00,00,000
Rising Main Pipeline	₹68,00,00,000
Total D&B Base Price (Estimate)	₹3,53,00,00,000
GST @ 18%	₹63,54,00,000
Total D&B Bid Value (Estimate, incl. GST)	₹4,16,54,00,000
10-Year O&M Fee (NPV @ 8% discount, Estimate)	₹89,00,00,000

I.1 O&M Year-wise Price Schedule Format

Bidders shall break down the quoted 10-Year O&M Fee on a year-wise basis in the format below, distinguishing the Fixed Fee component (manpower, routine maintenance) from the Variable Fee component (power, chemicals, consumables), for the Employer's cash-flow planning purposes.

O&M Year	Fixed Fee	Variable Fee	Total
Year 1	Fixed Fee (₹ Lakh) - to be quoted	Variable Fee (₹ Lakh) - to be quoted	Total (₹ Lakh) - to be quoted
Year 2	Fixed Fee (₹ Lakh) - to be quoted	Variable Fee (₹ Lakh) - to be quoted	Total (₹ Lakh) - to be quoted
Year 3	Fixed Fee (₹ Lakh) - to be quoted	Variable Fee (₹ Lakh) - to be quoted	Total (₹ Lakh) - to be quoted
Year 4	Fixed Fee (₹ Lakh) - to be quoted	Variable Fee (₹ Lakh) - to be quoted	Total (₹ Lakh) - to be quoted
Year 5	Fixed Fee (₹ Lakh) - to be quoted	Variable Fee (₹ Lakh) - to be quoted	Total (₹ Lakh) - to be quoted
Year 6	Fixed Fee (₹ Lakh) - to be quoted	Variable Fee (₹ Lakh) - to be quoted	Total (₹ Lakh) - to be quoted
Year 7	Fixed Fee (₹ Lakh) - to be quoted	Variable Fee (₹ Lakh) - to be quoted	Total (₹ Lakh) - to be quoted
Year 8	Fixed Fee (₹ Lakh) - to be quoted	Variable Fee (₹ Lakh) - to be quoted	Total (₹ Lakh) - to be quoted
Year 9	Fixed Fee (₹ Lakh) - to be quoted	Variable Fee (₹ Lakh) - to be quoted	Total (₹ Lakh) - to be quoted
Year 10	Fixed Fee (₹ Lakh) - to be quoted	Variable Fee (₹ Lakh) - to be quoted	Total (₹ Lakh) - to be quoted

I.2 Detailed D&B Bill of Quantities Format (Illustrative Line Items)

Bidders shall furnish a fully priced BOQ at the level of detail illustrated below (illustrative item descriptions and units shown; the actual BOQ to be issued as a separate priced-format spreadsheet at bid stage shall contain the complete, final item list and quantities).

Description	Unit
Site clearance, grading and levelling	sqm
Excavation for structures (ordinary soil)	cum

Description	Unit
Excavation for rising main trench	cum
Dewatering during excavation (as required)	LS
PCC M-10 for foundation bed	cum
RCC M-25 for water-retaining structures	cum
RCC M-20 for non-water-retaining structures	cum
Reinforcement steel (Fe-500D)	MT
Crystalline/membrane waterproofing	sqm
Structural steel (galvanized) - walkways/platforms	MT
Masonry works (blockwork)	cum
Plastering (including acid-resistant where specified)	sqm
Flooring (non-slip/chemical-resistant)	sqm
Roofing (RCC slab / PEB as applicable)	sqm
Doors, windows & ventilation louvers	sqm
Internal roads (bituminous)	sqm
Storm water drainage system	rmt
Boundary wall (RCC/masonry, 2.4m)	rmt
Landscaping	sqm
Chlorine safety structure (bunded, ventilated)	LS
Raw Water Pumping Station - civil	LS
Raw water pumps (vertical turbine, incl. motor)	No.
Raw water rising main pipeline (MS/DI)	rmt
Screening equipment (mechanically raked bar screens)	No.
Clariflocculator - mechanism (4 units)	No.
Tube settler modules	sqm
Rapid Gravity Filters - civil (16 beds)	LS
Filter media (sand + anthracite)	cum
Filter underdrain system (nozzle type)	sqm
Backwash pumps & blowers	Set
Chlorination system (chlorinators, ton container storage, scrubber)	LS
Gas leak detection & safety system	Set
Sludge thickener (2 Nos.)	No.
Sludge dewatering system (centrifuge/filter press)	No.
Sludge drying beds	sqm
Clear Water Reservoir - civil (16.7 ML)	LS
Clear Water Pumping Station - civil	LS
Clear water pumps (horizontal split case, incl. motor)	No.
Clear water rising main (8.5 km)	rmt
Surge protection system (surge vessels/relief valves)	Set
HT substation (11kV incoming, transformers)	LS
LT MCC panels (all process areas)	Set
DG sets (2x100% capacity)	No.
Power & control cabling (all areas)	rmt
Earthing & lightning protection system	LS
SCADA/PLC system (complete)	LS
Field instrumentation (flow/level/quality analyzers)	Set
Fiber optic communication backbone	rmt
Laboratory equipment (complete water testing lab)	LS
Administrative building - complete	sqm
Security cabin & CCTV system	LS
Fire fighting system (hydrants, extinguishers)	LS
HVAC system (control room, lab)	LS
Type testing & routine testing charges (all equipment)	LS
Third-party inspection charges	LS
Commissioning & Performance Guarantee Test support	LS
As-built documentation & O&M manuals	LS

Description	Unit
Training of Employer's O&M personnel	LS
Mandatory spares (2-year requirement)	LS
Insurance (CAR + TPL) for D&B phase	LS
Mobilization & demobilization	LS

PART J - Milestones & Payment Schedule

Milestone	Deliverable	Timeline (from Notice to Proceed)
M1	Mobilization, Insurance, Performance Security (D&B)	T + 15 days
M2	Detailed Design & Drawing Approval	T + 90 days
M3	Raw Water Pumping Station - Civil Complete	T + 240 days
M4	WTP Civil Structures - Complete	T + 420 days
M5	Mechanical & Electrical Equipment - Erection Complete	T + 540 days
M6	Rising Main - Laying & Testing Complete	T + 480 days
M7	SCADA Integration & Commissioning	T + 570 days
M8	30-Day Performance Guarantee Test Run	T + 570 to T + 600 days
M9	Provisional Acceptance & Start of O&M Period	T + 600 days (approx. 20 months)
M10	O&M Period	T + 600 days to T + 4250 days (10 years)

PART K - Standard Forms

Form K-1: Bid Submission Form

To be submitted on the bidder's letterhead, confirming acceptance of all Tender terms, the bid validity period, and enclosing the list of all documents submitted in the Technical Bid, duly signed by the authorized signatory holding a valid Power of Attorney.

Form K-2: Bid Security (EMD) Bank Guarantee Format

To be submitted on the format prescribed in Annexure L-3, issued by a Scheduled Commercial Bank, in favour of the Gujraat Water Supply & Sewerage Board, valid for the period specified in Part A, and confirming unconditional and irrevocable payment on first demand. The full text of the Bank Guarantee is reproduced below:

"WHEREAS [Bidder's Name] (hereinafter called "the Bidder") has submitted its bid dated [date] for the [name of contract] against Tender No. GWSSB/WTP-SURAT/2026/DBO-009 (hereinafter called "the Bid"). KNOW ALL PEOPLE by these presents that We [Bank Name] having our registered office at [address] (hereinafter called "the Bank") are bound unto the Gujraat Water Supply & Sewerage Board (hereinafter called "the Employer") in the sum of ₹2,00,00,000 (Rupees Two Crore only) for which payment well and truly to be made to the Employer, the Bank binds itself, its successors and assigns by these presents. THE CONDITIONS of this obligation are: (1) If the Bidder withdraws its Bid during the period of bid validity specified in the Form of Bid; or (2) If the Bidder, having been notified of the acceptance of its Bid by the Employer during the period of bid validity, fails or refuses to execute the Contract Agreement, or fails or refuses to furnish the Performance Security in accordance with the Instructions to Bidders; We undertake to pay to the Employer up to the above amount upon receipt of its first written demand, without the Employer having to substantiate its demand, provided that in its demand the Employer will note that the amount claimed by it is due to it, owing to the occurrence of one or both of the two conditions, specifying the occurred condition or conditions. This Guarantee will remain in force up to and including the date 30 (thirty) days after the deadline for submission of Bids as such deadline is stated in the Instructions to Bidders or as it may be extended by the Employer, notice of which extension(s) to the Bank is hereby waived. Any demand in respect of this Guarantee should reach the Bank not later than the above date."

Form K-3: Performance Security Bank Guarantee Format

To be submitted within 21 days of Notification of Award, in the format prescribed in Annexure L-4, covering both the D&B and O&M phases as separate instruments per SCC-6. The full text follows the same general structure as Form K-2 above, adapted to reference "due performance of the Contract" in place of "bid withdrawal", and shall be valid, in the case of the D&B-phase instrument, until 90 days beyond the expiry of the Defect Liability Period, and in the case of the O&M-phase instrument, until 90 days beyond the expiry of the 10-year O&M period, or as otherwise extended from time to time at the Employer's request.

Form K-4: Integrity Pact

To be executed between the Employer and the successful bidder at the time of Contract signing, in the standard Central Vigilance Commission (CVC) format, committing both parties to a corruption-free procurement and contract implementation process. Key commitments under the Integrity Pact include the following:

- The Employer commits that no employee of the Employer, personally or through any of its family members, will in connection with the tender for, or the execution of, the Contract, demand, take a promise for, or accept, for self or third person, any material or immaterial benefit which the person is not legally entitled to.
- The Bidder commits itself to take all measures necessary to prevent corrupt practices, unfair means and illegal activities during any stage of its bid or during any pre-contract or post-contract stage in order to secure the Contract or in furtherance to secure it and in particular commits itself not to give, promise or offer, directly or through intermediaries, any bribe, gift, consideration, reward, favour, or any material or immaterial benefit to any officer or employee of the Employer.
- The Bidder will not enter into any undisclosed agreement or understanding, whether formal or informal, with other Bidders with respect to prices, specifications, certifications, subsidiary contracts, or any other actions restrictive of competition.
- The Bidder will not commit any offence under the relevant Indian Penal Code / Prevention of Corruption Act provisions; further, the Bidder will not use improperly, for purposes of competition or personal gain, or pass on to others, any information provided by the Employer as part of the business relationship.
- If the Bidder or any employee of the Bidder, or any person acting on behalf of the Bidder, either directly or indirectly, is a relative of any of the officers of the Employer, or alternatively, if any relative of an officer of the Employer has financial interest/stake in the Bidder's firm, the same shall be disclosed by the Bidder at the time of filing the Tender.
- The Employer will exclude the Bidder from future tenders for a period as decided by the Employer (which may be up to a maximum of 5 years) if the Bidder is found to have breached the Integrity Pact, in addition to any other legal action that may be initiated.

Form K-5: Letter of Authorization (Power of Attorney)

To be submitted authorizing the signatory of the bid to commit the bidder (or, in case of a JV, all JV members) to the terms of the bid and the resulting Contract. The Power of Attorney shall be executed on non-judicial stamp paper of appropriate value as per the Gujraat Stamp Act, and shall be notarized.

Form K-6: Joint Venture (JV) Agreement Format (if applicable)

Where the bidder is a Joint Venture, a notarized JV Agreement shall be submitted in the format prescribed, clearly defining the roles, responsibilities, and joint & several liability of each JV member for both the D&B and O&M phases of the Contract, including the specific scope of work to be undertaken by each member and the percentage equity/profit-sharing arrangement between them.

Form K-7: Self-Declaration of Eligibility

A self-declaration, on the bidder's letterhead, confirming that the bidder meets all Eligibility Criteria set out in Part F, and that all information and documents submitted are true and correct, and that the bidder undertakes to inform the Employer promptly of any change in circumstances that might affect the accuracy or validity of this declaration at any time prior to Contract award.

PART L-A - GWSSB Standard Specifications for Civil Works (Extract)

The following extract from GWSSB's Standard Specifications for Civil Works shall apply to this Contract, in addition to the project-specific Technical Specifications set out in Part E.

CS-1: Earthwork & Site Grading

All earthwork for site grading, excavation for structures, and backfilling shall be carried out in accordance with IS 1200 (Part 1) and the approved grading plan. Excavated material suitable for reuse shall be stockpiled separately from unsuitable material; surplus/unsuitable material shall be disposed of at a location approved by the Employer, at the Contractor's cost.

All materials and workmanship under this Clause shall be subject to inspection and testing by the Employer's Engineer-in-Charge or nominated Third-Party Inspection Agency at the frequency and method specified in the project-specific Quality Assurance Plan to be submitted by the Contractor and approved by the Employer within 30 days of the Notice to Proceed.

CS-2: Concrete Works

All RCC works shall conform to IS 456:2000 and shall use a minimum grade of M-25 for water-retaining structures and M-20 for non-water-retaining structures, unless a higher grade is specified in the structural design. Water-retaining structures shall additionally satisfy the crack-width and durability requirements of IS 3370 (Parts 1-4).

All materials and workmanship under this Clause shall be subject to inspection and testing by the Employer's Engineer-in-Charge or nominated Third-Party Inspection Agency at the frequency and method specified in the project-specific Quality Assurance Plan to be submitted by the Contractor and approved by the Employer within 30 days of the Notice to Proceed.

CS-3: Reinforcement Steel

Reinforcement steel shall be Fe-500D or higher grade TMT bars conforming to IS 1786, sourced from a primary steel producer on the Approved Makes List (Annexure L-5). Cover to reinforcement in water-retaining structures shall be a minimum of 45mm, or as per the structural drawings, whichever is greater.

All materials and workmanship under this Clause shall be subject to inspection and testing by the Employer's Engineer-in-Charge or nominated Third-Party Inspection Agency at the frequency and method specified in the project-specific Quality Assurance Plan to be submitted by the Contractor and approved by the Employer within 30 days of the Notice to Proceed.

CS-4: Waterproofing

All water-retaining structures (clariflocculators, filter beds, clear water reservoir) shall be provided with an approved crystalline or membrane waterproofing system in addition to dense, well-compacted concrete, and shall be subject to a mandatory hydrostatic (water) test prior to backfilling and prior to commissioning.

All materials and workmanship under this Clause shall be subject to inspection and testing by the Employer's Engineer-in-Charge or nominated Third-Party Inspection Agency at the frequency and method specified in the project-specific Quality Assurance Plan to be submitted by the Contractor and approved by the Employer within 30 days of the Notice to Proceed.

CS-5: Structural Steel

Structural steel for walkways, platforms, and handrails shall conform to IS 2062, hot-dip galvanized as per IS 4759 to a minimum coating thickness of 86 microns, with all site connections bolted rather than welded to facilitate future maintenance and replacement.

All materials and workmanship under this Clause shall be subject to inspection and testing by the Employer's Engineer-in-Charge or nominated Third-Party Inspection Agency at the frequency and method specified in the project-specific Quality Assurance Plan to be submitted by the Contractor and approved by the Employer within 30 days of the Notice to Proceed.

CS-6: Masonry & Plastering

Masonry works for non-structural walls shall use solid concrete blocks or fly-ash bricks conforming to relevant IS codes; internal plastering in chemical dosing and chlorination areas shall be acid-resistant / chemical-resistant as appropriate to the specific chemical exposure.

All materials and workmanship under this Clause shall be subject to inspection and testing by the Employer's Engineer-in-Charge or nominated Third-Party Inspection Agency at the frequency and method specified in the project-specific Quality Assurance Plan to be submitted by the Contractor and approved by the Employer within 30 days of the Notice to Proceed.

CS-7: Flooring

Flooring in operational areas (pump halls, filter gallery, chemical house) shall be non-slip, chemical-resistant (where applicable), and designed for the anticipated foot and equipment traffic loads, including provision for equipment removal/replacement routes.

All materials and workmanship under this Clause shall be subject to inspection and testing by the Employer's Engineer-in-Charge or nominated Third-Party Inspection Agency at the frequency and method specified in the project-specific Quality Assurance Plan to be submitted by the Contractor and approved by the Employer within 30 days of the Notice to Proceed.

CS-8: Roofing

Roofing over process structures shall be designed to prevent rainwater ingress into open water-retaining structures while allowing adequate ventilation; RCC slab roofing is preferred for the WTP main structures, with pre-engineered steel structures permissible for ancillary buildings subject to Employer approval.

All materials and workmanship under this Clause shall be subject to inspection and testing by the Employer's Engineer-in-Charge or nominated Third-Party Inspection Agency at the frequency and

method specified in the project-specific Quality Assurance Plan to be submitted by the Contractor and approved by the Employer within 30 days of the Notice to Proceed.

CS-9: Doors, Windows & Ventilation

All operational buildings shall be provided with adequate natural and/or mechanical ventilation, particularly the chlorine storage/dosing room, which shall have forced ventilation with discharge through the gas scrubber system per CS-14 (Chlorine Safety).

All materials and workmanship under this Clause shall be subject to inspection and testing by the Employer's Engineer-in-Charge or nominated Third-Party Inspection Agency at the frequency and method specified in the project-specific Quality Assurance Plan to be submitted by the Contractor and approved by the Employer within 30 days of the Notice to Proceed.

CS-10: Roads & Pavements

Internal roads shall be designed for the anticipated heavy vehicle traffic (chemical tankers, chlorine tonners, equipment transport) with a minimum 2-coat bituminous surface over a graded base course, or RCC pavement in high-load areas (chemical unloading bay).

All materials and workmanship under this Clause shall be subject to inspection and testing by the Employer's Engineer-in-Charge or nominated Third-Party Inspection Agency at the frequency and method specified in the project-specific Quality Assurance Plan to be submitted by the Contractor and approved by the Employer within 30 days of the Notice to Proceed.

CS-11: Storm Water Drainage

A comprehensive storm water drainage system shall be provided for the entire site, sized for the local design rainfall intensity, discharging to an approved outfall point, with provision for silt traps at all discharge points.

All materials and workmanship under this Clause shall be subject to inspection and testing by the Employer's Engineer-in-Charge or nominated Third-Party Inspection Agency at the frequency and method specified in the project-specific Quality Assurance Plan to be submitted by the Contractor and approved by the Employer within 30 days of the Notice to Proceed.

CS-12: Boundary Wall & Security

The site perimeter shall be enclosed by an RCC/masonry compound wall of minimum 2.4m height with anti-climb provisions, along with a manned main gate, security cabin, and CCTV coverage of all critical access points and process areas.

All materials and workmanship under this Clause shall be subject to inspection and testing by the Employer's Engineer-in-Charge or nominated Third-Party Inspection Agency at the frequency and method specified in the project-specific Quality Assurance Plan to be submitted by the Contractor and approved by the Employer within 30 days of the Notice to Proceed.

CS-13: Landscaping

A minimum of 10% of the site area (excluding process structures) shall be landscaped with native, low-water-demand plant species, in accordance with the Employer's green-building guidelines for utility infrastructure.

All materials and workmanship under this Clause shall be subject to inspection and testing by the Employer's Engineer-in-Charge or nominated Third-Party Inspection Agency at the frequency and method specified in the project-specific Quality Assurance Plan to be submitted by the Contractor and approved by the Employer within 30 days of the Notice to Proceed.

CS-14: Chlorine Safety Structures

The chlorine storage and dosing area shall be designed as a segregated, bunded structure with forced-draft ventilation, gas leak detection interlocked with the scrubber system, and an emergency shower/eyewash station, in full compliance with CPCB's Manufacture, Storage and Import of Hazardous Chemicals Rules.

All materials and workmanship under this Clause shall be subject to inspection and testing by the Employer's Engineer-in-Charge or nominated Third-Party Inspection Agency at the frequency and method specified in the project-specific Quality Assurance Plan to be submitted by the Contractor and approved by the Employer within 30 days of the Notice to Proceed.

CS-15: Seismic Design

All structures shall be designed for Seismic Zone III (as applicable to Surat) in accordance with IS 1893 (Part 1 and Part 2 for liquid-retaining structures), with importance factor appropriate to critical public infrastructure.

All materials and workmanship under this Clause shall be subject to inspection and testing by the Employer's Engineer-in-Charge or nominated Third-Party Inspection Agency at the frequency and method specified in the project-specific Quality Assurance Plan to be submitted by the Contractor and approved by the Employer within 30 days of the Notice to Proceed.

PART L-B - GWSSB Standard Specifications for Mechanical & Electrical Works (Extract)

The following extract from GWSSB's Standard Specifications for Mechanical & Electrical Works shall apply to this Contract, in addition to the project-specific Technical Specifications set out in Part E.

MS-1: Pump Selection & Testing

All pumps shall be selected to operate at or near their best efficiency point (BEP) across the anticipated operating range, and shall be factory performance-tested in accordance with IS 9137 / Hydraulic Institute Standards prior to dispatch, with test certificates furnished to the Employer.

Manufacturer's type test certificates and, where specified in the Approved Vendor List / Approved Makes List (Annexure L-5), OEM authorization letters shall be furnished by the Contractor at the time of drawing/GTP submission for this equipment category, prior to release of manufacturing.

MS-2: Motors

All motors shall be IE3 (premium efficiency) or better, TEFC enclosure, suitable for the site's ambient and humidity conditions, with Class F insulation and Class B temperature rise, sized with a minimum 15% service factor margin over the driven equipment's maximum power demand.

Manufacturer's type test certificates and, where specified in the Approved Vendor List / Approved Makes List (Annexure L-5), OEM authorization letters shall be furnished by the Contractor at the time of drawing/GTP submission for this equipment category, prior to release of manufacturing.

MS-3: Piping & Valves

Process piping shall be MS/DI/uPVC as appropriate to the service and pressure rating; all isolation valves shall be resilient-seated gate/butterfly valves; check valves at pump discharge shall be non-slam type to prevent water hammer.

Manufacturer's type test certificates and, where specified in the Approved Vendor List / Approved Makes List (Annexure L-5), OEM authorization letters shall be furnished by the Contractor at the time of drawing/GTP submission for this equipment category, prior to release of manufacturing.

MS-4: Cranes & Lifting Equipment

EOT cranes shall be provided in the pump halls and mechanical equipment rooms sized for the heaviest single component requiring removal (typically the pump or motor), in accordance with IS 3177.

Manufacturer's type test certificates and, where specified in the Approved Vendor List / Approved Makes List (Annexure L-5), OEM authorization letters shall be furnished by the Contractor at the time of drawing/GTP submission for this equipment category, prior to release of manufacturing.

MS-5: HT/LT Electrical Systems

The WTP shall be provided with a dedicated HT substation (11kV incoming) with dual-feeder redundancy from the Discom, stepped down via dry-type or oil-filled transformers to LT distribution for all plant loads, with adequate power factor correction capacitor banks.

Manufacturer's type test certificates and, where specified in the Approved Vendor List / Approved Makes List (Annexure L-5), OEM authorization letters shall be furnished by the Contractor at the time of drawing/GTP submission for this equipment category, prior to release of manufacturing.

MS-6: DG Backup

A minimum of 2 x 100% capacity DG sets shall be provided to back up all critical loads (raw water pumps, clariflocculator drives, filter backwash system, clear water pumps, chlorination, SCADA) in the event of grid power failure, with automatic changeover.

Manufacturer's type test certificates and, where specified in the Approved Vendor List / Approved Makes List (Annexure L-5), OEM authorization letters shall be furnished by the Contractor at the time of drawing/GTP submission for this equipment category, prior to release of manufacturing.

MS-7: Motor Control Centres (MCC)

MCCs shall be provided for each process area, with soft-starters/VFDs for large motors as specified in Part E, and shall be integrated with the SCADA system for remote start/stop, status monitoring, and fault annunciation.

Manufacturer's type test certificates and, where specified in the Approved Vendor List / Approved Makes List (Annexure L-5), OEM authorization letters shall be furnished by the Contractor at the time of drawing/GTP submission for this equipment category, prior to release of manufacturing.

MS-8: Earthing & Lightning Protection

A comprehensive earthing grid shall be provided per IS 3043, with lightning protection for all elevated structures (CWR, elevated tanks if any, chemical storage) per IS 2309.

Manufacturer's type test certificates and, where specified in the Approved Vendor List / Approved Makes List (Annexure L-5), OEM authorization letters shall be furnished by the Contractor at the time of drawing/GTP submission for this equipment category, prior to release of manufacturing.

MS-9: Instrumentation Calibration

All field instruments (flow meters, level transmitters, analyzers) shall be factory-calibrated and shall additionally be field-calibrated against reference standards at commissioning and at intervals not exceeding 6 months during the O&M period.

Manufacturer's type test certificates and, where specified in the Approved Vendor List / Approved Makes List (Annexure L-5), OEM authorization letters shall be furnished by the Contractor at the time of drawing/GTP submission for this equipment category, prior to release of manufacturing.

MS-10: Spare Parts & Special Tools

The Contractor shall supply a comprehensive list of mandatory spares (2-year O&M requirement) and special tools/tackles required for maintenance of all major equipment, as detailed in the technical proposal.

Manufacturer's type test certificates and, where specified in the Approved Vendor List / Approved Makes List (Annexure L-5), OEM authorization letters shall be furnished by the Contractor at the time of drawing/GTP submission for this equipment category, prior to release of manufacturing.

Annexure L-2 - Raw Water Characteristics Data (36-Month Historical Record)

The following table presents the monthly raw water quality data recorded at the existing Tapi River intake well over the preceding 36 months, for bidders' reference in process design. Bidders shall independently verify this data through their own site investigation and shall not hold the Employer responsible for any variation encountered during actual operation.

Month	Turbidity (NTU)	pH	Alkalinity (mg/l)	Total Hardness (mg/l)
Y1-Jan	39	7.7	87	174
Y1-Feb	37	7.6	85	238
Y1-Mar	45	7.5	97	228
Y1-Apr	49	7.1	110	174
Y1-May	50	7.2	114	199
Y1-Jun	576	7.1	136	247
Y1-Jul	521	7.5	137	169
Y1-Aug	819	7.7	120	178
Y1-Sep	570	8.2	107	173
Y1-Oct	43	7.4	89	227
Y1-Nov	31	7.4	85	243
Y1-Dec	24	8.1	108	174
Y2-Jan	46	7.5	126	215
Y2-Feb	18	7.3	107	159
Y2-Mar	13	7.5	121	215
Y2-Apr	67	7.4	106	200
Y2-May	36	8.0	107	194
Y2-Jun	664	7.3	125	163
Y2-Jul	670	7.9	132	247
Y2-Aug	1327	8.0	125	214
Y2-Sep	955	7.0	99	186
Y2-Oct	23	7.5	121	224
Y2-Nov	26	8.1	80	186
Y2-Dec	45	7.5	85	249
Y3-Jan	37	8.0	115	197
Y3-Feb	18	8.0	134	176
Y3-Mar	11	7.2	122	220
Y3-Apr	50	8.0	94	233
Y3-May	20	7.0	85	152
Y3-Jun	937	8.0	131	165
Y3-Jul	894	7.1	97	199
Y3-Aug	464	7.4	94	180
Y3-Sep	1191	7.2	90	150

Month	Turbidity (NTU)	pH	Alkalinity (mg/l)	Total Hardness (mg/l)
Y3-Oct	47	7.1	104	161
Y3-Nov	39	7.5	81	226
Y3-Dec	37	7.9	100	243

Annexure L-5 - Approved Makes List for Bulk Materials & Equipment

The following Approved Makes List shall apply for bulk materials and standard equipment items. Major process equipment (pumps, clariflocculator mechanisms, filter systems, chlorination systems) shall be sourced from Original Equipment Manufacturers approved separately through the Approved Vendor List (AVL) empanelment process referenced in Part F.

Material / Equipment Category	Approved Makes (any one)
Cement (OPC 43/53 Grade)	Ultratech / Ambuja / ACC
TMT Reinforcement Steel	TATA Tiscon / SAIL / JSW
Structural Steel Sections	SAIL / JSPL / Jindal
Waterproofing System (crystalline)	Fosroc / Sika / BASF
MS/DI Pipes (rising main)	Jindal / Electrosteel / Tata Ductura
Butterfly Valves	Kirloskar Brothers / L&T Valves / IVC
Gate Valves	Kirloskar Brothers / Leader Valves / IVC
Non-return Valves	Kirloskar Brothers / L&T Valves
Vertical Turbine Pumps (Raw Water)	Kirloskar Brothers / WPIL / Beacon Weir
Horizontal Split Case Pumps (Clear Water)	Kirloskar Brothers / WPIL / KSB
Motors	ABB / Siemens / Crompton Greaves / Kirloskar Electric
VFDs	ABB / Siemens / Danfoss / Schneider
HT Switchgear	Siemens / ABB / Schneider
LT Switchgear / MCC	L&T / Siemens / ABB
Transformers	Voltamp / Kirloskar Electric / Crompton Greaves
DG Sets	Cummins / Kirloskar / Mahindra Powerol
Cables	Polycab / KEI / Havells
Chlorinators	Wallace & Tiernan / Advance Water Treatment / Prominent
Flow Meters (Electromagnetic)	Krohne / Endress+Hauser / ABB
Level Transmitters	Endress+Hauser / Vega / Siemens
Turbidity/Chlorine Analyzers	Hach / Endress+Hauser / Emerson
PLC/SCADA Hardware	Siemens / Rockwell / Schneider
Sluice Gates	Bharat Heavy Plate & Vessels / Aqua Gates India
Filter Nozzles	Fairfield / Enviro Care / Aqua Design
Sludge Dewatering Centrifuge	Alfa Laval / Andritz / Flottweg
Filter Press	Diemme / JVS Engineering
EOT Cranes	ACE / Anupam / Hercules Hoists
Gas Detectors (chlorine)	Honeywell / Draeger / MSA
Fire Fighting Equipment	Ceasefire / Safex / Minimax
Paints (structural)	Asian Paints / Berger / Jotun
Water-stops (construction joints)	Fosroc / Sika / CCL
HDPE Pipes (drainage/utility)	Supreme / Finolex
Cable Trays	Legrand / Indiana / Slotco
UPS Systems	APC / Emerson / Delta
Batteries (VRLA)	Exide / Amara Raja / HBL
Lifts/Elevators (admin building)	Otis / Kone / Schindler
HVAC Equipment	Voltas / Blue Star / Daikin
Lab Equipment (general)	Thermo Fisher / Eutech / Systronics
CCTV Systems	Hikvision / CP Plus / Bosch
Street/Yard Lighting	Philips / Bajaj / Crompton

Annexure L-7 - CPHEEO Manual Design Criteria (Reference Extract)

The following design criteria, extracted from the Central Public Health & Environmental Engineering Organisation (CPHEEO) Manual on Water Supply and Treatment, shall guide the detailed design and shall be read together with the project-specific parameters in Part E, which shall govern in case of any conflict.

Design Parameter	CPHEEO Recommended Range
Rapid Mix Detention Time	30 - 60 seconds
Flocculation Detention Time	20 - 30 minutes
Flocculation Velocity Gradient (G)	10 - 75 sec ⁻¹
Clariflocculator Surface Loading	1.0 - 1.5 m ³ /m ² /hr
Clarifier Weir Loading Rate	≤ 300 m ³ /m/day
Rapid Sand Filter Rate	5 - 7.5 m ³ /m ² /hr
Filter Media Depth (sand)	60 - 90 cm
Filter Media Depth (anthracite, dual media)	30 - 45 cm
Filter Backwash Rate	36 - 44 m ³ /m ² /hr
Air Scour Rate	0.6 - 0.9 m ³ /m ² /min
Chlorine Contact Time (minimum)	30 minutes at peak flow
Clear Water Reservoir Storage	3 - 4 hours of average demand (minimum)
Raw Water Pumping - Velocity in Rising Main	1.0 - 2.0 m/s (economical velocity range)
Sludge Production Estimate	1.5 - 2.5% of raw water volume, dependent on turbidity
Alum Dose Range	10 - 60 mg/l (turbidity dependent)
Design Population Growth Rate	As per Census/UDPFI guidelines, project-specific
Per Capita Water Demand (design)	135 - 150 lpcd, as per CPHEEO norms for Class I cities
Non-Revenue Water Allowance	15% (design allowance for distribution losses)
Fire Demand Allowance	As per National Building Code provisions
Standby/Redundancy - Pumps	Minimum N+1 or N+2 as per criticality of duty

PART L - Annexures

- Annexure L-1: Site Data Report (topography, geotechnical summary, existing utilities)
- Annexure L-2: Raw Water Characteristics Data (3-year historical, monthly)
- Annexure L-3: Bid Security (EMD) Bank Guarantee - Detailed Format
- Annexure L-4: Performance Security Bank Guarantee - Detailed Format
- Annexure L-5: List of Approved Makes for Bulk Materials & Equipment
- Annexure L-6: Drawing List (preliminary GA drawings issued with Tender)
- Annexure L-7: CPHEEO Manual Design Criteria - Reference Extract
- Annexure L-8: GWSSB Standard Specifications for Civil Works
- Annexure L-9: GWSSB Standard Specifications for Mechanical/Electrical Works
- Annexure L-10: Format for Price Bid Submission (D&B and O&M Schedules)

Annexure L-1 - Site Data Report

The Site for the proposed WTP is located at Village Kosad, on the eastern outskirts of Surat city, approximately 12 km from the existing GWSSB intake well on the Tapi River. The site comprises approximately 18 hectares of land, largely flat with a gentle slope towards the south-east, currently under GWSSB's ownership and free from encroachment as of the date of issue of this Tender Document.

Preliminary geotechnical investigation carried out by the Employer indicates a predominantly alluvial soil profile, with a safe bearing capacity in the range of 12-18 t/sqm at a founding depth of 1.5-2.5m below existing ground level, though the Contractor shall be required to carry out its own detailed geotechnical investigation, including bore holes at all major structure locations, prior to finalizing foundation design. The groundwater table was recorded at approximately 3.5m below ground level during the pre-monsoon survey and may rise closer to the surface during the monsoon season, requiring dewatering provisions during construction of below-ground structures.

The site is accessible via an existing 12m-wide approach road connecting to the Surat-Kosad state highway, approximately 800m from the site boundary; the Contractor shall be responsible for any strengthening of this approach road required to support heavy equipment and material transport during construction, subject to the Employer's approval of the scope and cost of such strengthening as a reimbursable item.

Grid power is available in the vicinity via an existing 11kV feeder approximately 1.2 km from the site boundary; the Contractor shall coordinate with the local Discom for the necessary HT connection augmentation and shall bear the cost of extending the feeder to the site boundary, with connection charges reimbursed by the Employer as a pass-through cost under the Contract.

No forest land, wildlife sanctuary, or other environmentally sensitive area is located within or adjacent to the project site; however, the site lies within 500m of the Tapi River floodplain, and the Contractor's design shall account for the 100-year flood level as provided in the hydrological data available with the Employer's Irrigation Department, ensuring all critical electrical and control equipment is located above this flood level.

Annexure L-6 - Drawing List (Preliminary GA Drawings Issued with Tender)

The following preliminary General Arrangement (GA) drawings are issued with this Tender Document for the bidder's reference in preparing its technical proposal. These are indicative only; the Contractor shall be responsible for detailed engineering and preparation of Good-for-Construction drawings as set out in Part D.1.

S.No.	Drawing Title
1	Overall Site Layout Plan
2	Raw Water Pumping Station - GA Plan & Section
3	WTP Process Flow Diagram (PFD)
4	WTP Overall Layout - Ground Floor Plan
5	Clariflocculator - GA Plan & Section
6	Rapid Gravity Filter - GA Plan & Section
7	Chlorination Building - GA Plan & Section
8	Sludge Handling Area - GA Plan & Section
9	Clear Water Reservoir - GA Plan & Section
10	Clear Water Pumping Station - GA Plan & Section
11	Administrative Building - Floor Plans (all levels)
12	Laboratory Building - Floor Plan
13	Electrical Substation - GA Plan & Section
14	Boundary Wall & Security - Layout Plan
15	Internal Road Network - Layout Plan
16	Storm Water Drainage - Layout Plan
17	Rising Main - Alignment Plan (Sheet 1 of 3)
18	Rising Main - Alignment Plan (Sheet 2 of 3)
19	Rising Main - Alignment Plan (Sheet 3 of 3)
20	Rising Main - Longitudinal Section
21	Hydraulic Profile - Complete Treatment Train
22	Single Line Diagram - Electrical System (Preliminary)
23	SCADA Architecture Diagram (Preliminary)
24	Landscaping & Green Area Plan
25	Site Grading & Levelling Plan
26	Existing Utilities Plan (as available with Employer)
27	Topographic Survey Plan
28	Soil Investigation - Bore Hole Location Plan
29	Fire Fighting System - Layout Plan (Preliminary)
30	Ancillary Buildings - Consolidated Layout

Annexure L-9 - Glossary of Terms

Term	Definition
AMRUT	Atal Mission for Rejuvenation and Urban Transformation, a Government of India urban infrastructure funding scheme
BOQ	Bill of Quantities
CPHEEO	Central Public Health & Environmental Engineering Organisation, Ministry of Housing and Urban Affairs, Government of India
CWR	Clear Water Reservoir
D&B	Design-Build (referring to the capital works phase of the Contract)
DBO	Design, Build and Operate
DLP	Defect Liability Period
EMD	Earnest Money Deposit
GCC	General Conditions of Contract

Term	Definition
GPCB	Gujarat Pollution Control Board
GWSSB	Gujarat Water Supply & Sewerage Board (the Employer)
HT/LT	High Tension / Low Tension (electrical supply classification)
ITB	Instructions to Bidders
LPS/LPCD	Litres Per Second / Litres Per Capita per Day
MLD	Million Litres per Day
NPV	Net Present Value
NTU	Nephelometric Turbidity Unit
O&M	Operation & Maintenance
PLC	Programmable Logic Controller
RGF	Rapid Gravity Filter
SCADA	Supervisory Control and Data Acquisition
SCC	Special Conditions of Contract
TPIA	Third-Party Inspection Agency
VFD	Variable Frequency Drive
WPI	Wholesale Price Index
WTP	Water Treatment Plant

Annexure L-10 - List of Applicable Indian Standards & Codes

Code / Reference	Title / Applicability
IS 456:2000	Code of Practice for Plain and Reinforced Concrete
IS 1893 (Part 1 & 2)	Criteria for Earthquake Resistant Design of Structures
IS 3370 (Parts 1-4)	Code of Practice for Concrete Structures for Storage of Liquids
IS 1200	Method of Measurement of Building and Civil Engineering Works
IS 1786	High Strength Deformed Steel Bars for Concrete Reinforcement
IS 2062	Steel for General Structural Purposes
IS 4759	Hot-Dip Zinc Coatings on Structural Steel
IS 3177	Code of Practice for Electric Overhead Travelling Cranes
IS 3043	Code of Practice for Earthing
IS 2309	Code of Practice for Protection of Buildings and Allied Structures Against Lightning
IS 9137	Code for Acceptance Test for Centrifugal, Mixed Flow and Axial Pumps
IS 10500:2012	Drinking Water - Specification
IS 2065	Code of Practice for Water Supply in Buildings
IS 5613	Code of Practice for Design, Installation and Maintenance of Overhead Power Lines
IS 1554	PVC Insulated (Heavy Duty) Electric Cables
IS 8034	Guide for Selection, Installation and Maintenance of Vertical Turbine Pumps
IS 4984	High Density Polyethylene Pipes for Water Supply
IS 8329	Centrifugally Cast (Spun) Ductile Iron Pressure Pipes
CPHEEO Manual on Water Supply and Treatment	Reference design manual, Ministry of Housing and Urban Affairs
CPCB Guidelines for Water Treatment Plant Sludge Disposal	Environmental compliance guidelines
National Building Code of India (NBC)	Reference for fire safety, building services
Factories Act, 1948	Applicable for worker safety and welfare during construction and O&M
Building and Other Construction Workers Act, 1996	Applicable for construction workforce welfare
Gujarat Pollution Control Board (Consent to Establish/Operate) Regulations	Environmental statutory compliance
Manufacture, Storage and Import of	Applicable to chlorine gas storage and handling

Code / Reference	Title / Applicability
Hazardous Chemicals Rules, 1989	

Annexure L-11 - Illustrative Pre-Bid Queries & Clarifications

The following is an illustrative (non-exhaustive) set of the type of queries typically raised by prospective bidders during the pre-bid clarification period, provided here for guidance. Actual queries and responses for this Tender will be published as a formal Addendum following the Pre-Bid Meeting specified in Part A.

Illustrative Query	Illustrative Response
Can the D&B and O&M Performance Securities be combined into a single instrument?	No; per SCC-6, separate instruments are required for the D&B and O&M phases respectively.
Is a Joint Venture of more than 2 members permitted?	No; per ITB-3, JVs are limited to a maximum of 2 members.
What is the basis for the O&M price escalation cap?	WPI-linked, capped at 7% per annum, per SCC-9, applied from Year 2 of the O&M period onward.
Will the Employer provide the land free of cost?	Yes; per SCC-2, the site is already in the Employer's possession and will be handed over free from encumbrances.
Is power during construction to be arranged by the Contractor?	Yes, at the Contractor's cost, though grid augmentation charges for the O&M-phase HT connection are reimbursable per Annexure L-1.
Can sodium hypochlorite be proposed instead of gas chlorination?	Yes, as an explicitly declared deviation per D.7, subject to the Employer's technical evaluation of equivalence.
What is the discount rate for NPV computation of the O&M price for evaluation?	8% per annum, as specified in Part G (Evaluation Criteria).
Is there a minimum local employment requirement during O&M?	No specific quota is mandated in this Tender; however, bidders are encouraged to prioritize local hiring for unskilled/semi-skilled roles.
Can the bidder propose an alternate filtration technology (e.g., membrane filtration)?	Any alternate technology proposal must be submitted as a clearly declared deviation with full technical and cost justification; the base design intent (RGF) as per Part E.4 remains the Employer's preferred and default technology.
What third-party testing agency will be used for treated water quality monitoring during O&M?	To be mutually agreed between the Employer and Contractor from a panel of NABL-accredited laboratories, at Contract signing.